



School of Artificial Intelligence
Jilin University

第五章 动态规划

吉林大学人工智能学院

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- 5.1 0-1背包问题
- 5.2 动态规划主要思想
- 5.3 再论背包问题
- 5.4 可靠性设计
- 5.5 最短路径问题
- 5.6 再论货郎担问题
- 5.7 最长公共子序列
- 5.8 编辑距离问题
- 5.9 最优二分检索树
- 5.10 矩阵链乘积问题

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- **5.7 最长公共子序列**
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- 5.9 最优二分检索树
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- 子序列
 - 将给定序列中零个或多个元素（如字符）去掉后所得结果

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 - 将给定序列中零个或多个元素（如字符）去掉后所得结果
- 示例
 - 给定序列 X

X	A	B	C	B	D	A	B
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问题背景：子序列

- 子序列
 - 将给定序列中零个或多个元素（如字符）去掉后所得结果
- 示例
 - 给定序列 X

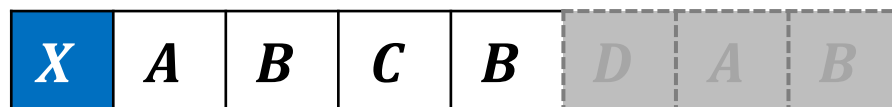
X	A	B	C	B	D	A	B
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- X 的子序列

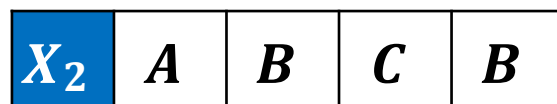
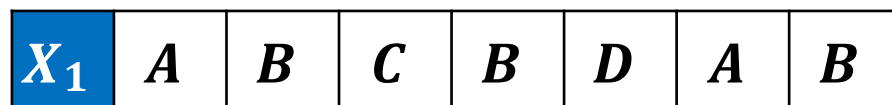
X_1	A	B	C	B	D	A	B
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问题背景：子序列

- 子序列
 - 将给定序列中零个或多个元素（如字符）去掉后所得结果
- 示例
 - 给定序列 X

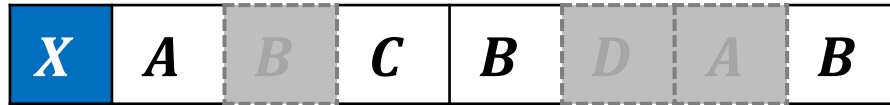


- X 的子序列

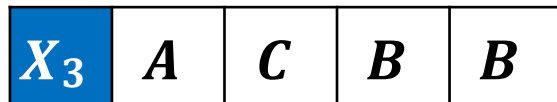
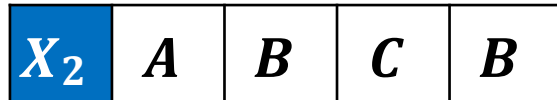
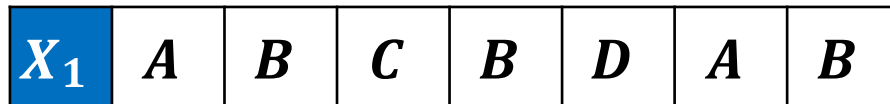


问题背景：子序列

- 子序列
 - 将给定序列中零个或多个元素（如字符）去掉后所得结果
- 示例
 - 给定序列 X



- X 的子序列



问题背景：公共子序列

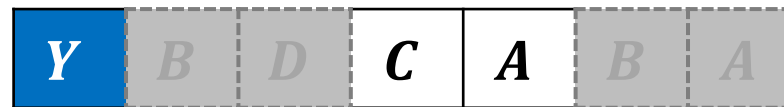
- 给定两个序列 X 和 Y

X	A	B	C	B	D	A	B
-----	-----	-----	-----	-----	-----	-----	-----

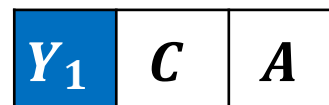
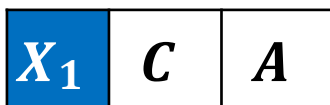
Y	B	D	C	A	B	A
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问题背景：公共子序列

- 给定两个序列 X 和 Y



- 公共子序列示例

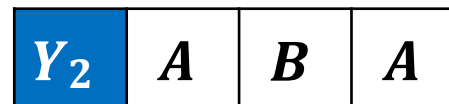
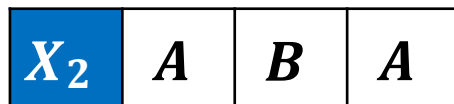
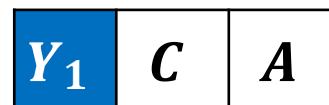
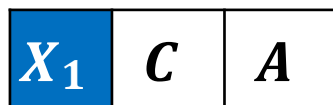


问题背景：公共子序列

- 给定两个序列 X 和 Y



- 公共子序列示例



问题背景：公共子序列

- 给定两个序列 X 和 Y

X	A	B	C	B	D	A	B
-----	-----	-----	-----	-----	-----	-----	-----

Y	B	D	C	A	B	A
-----	-----	-----	-----	-----	-----	-----

- 公共子序列示例

X_1	C	A
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Y_1	C	A
-------	-----	-----

X_2	A	B	A
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Y_2	A	B	A
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X_3	B	C	A	B
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Y_3	B	C	A	B
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问题背景：公共子序列

- 给定两个序列 X 和 Y

X	A	B	C	B	D	A	B
-----	-----	-----	-----	-----	-----	-----	-----

Y	B	D	C	A	B	A
-----	-----	-----	-----	-----	-----	-----

- 公共子序列示例

X_1	C	A
-------	-----	-----

Y_1	C	A
-------	-----	-----

X_2	A	B	A
-------	-----	-----	-----

Y_2	A	B	A
-------	-----	-----	-----

X_3	B	C	A	B
-------	-----	-----	-----	-----

Y_3	B	C	A	B
-------	-----	-----	-----	-----

问题：如何求两个给定序列的最长公共子序列？

- 序列 $X = \langle x_1, x_2, \dots, x_n \rangle$ 和序列 $Y = \langle y_1, y_2, \dots, y_m \rangle$
- 求解一个公共子序列 $Z = \langle z_1, z_2, \dots, z_l \rangle$, 令

$$\max |Z|$$

优化目标

$$s. t. \langle z_1, z_2, \dots, z_l \rangle = \langle x_{i_1}, x_{i_2}, \dots, x_{i_l} \rangle = \langle y_{j_1}, y_{j_2}, \dots, y_{j_l} \rangle$$

$$(1 \leq i_1 < i_2, \dots, i_l \leq n; 1 \leq j_1 < j_2, \dots, j_l \leq m)$$

约束条件

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>
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枚举并检查长度为1的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

枚举并检查长度为1的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>X</i>	<i>C</i>
----------	----------

枚举并检查长度为1的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>X</i>	<i>C</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>X</i>	<i>D</i>
----------	----------

<i>X</i>	<i>A</i>
----------	----------

<i>X</i>	<i>B</i>
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枚举并检查长度为1的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>
----------	----------

<i>Y</i>	<i>B</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>D</i>
----------	----------

<i>X</i>	<i>C</i>
----------	----------

<i>Y</i>	<i>C</i>
----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>A</i>
----------	----------

<i>X</i>	<i>D</i>
----------	----------

<i>Y</i>	<i>B</i>
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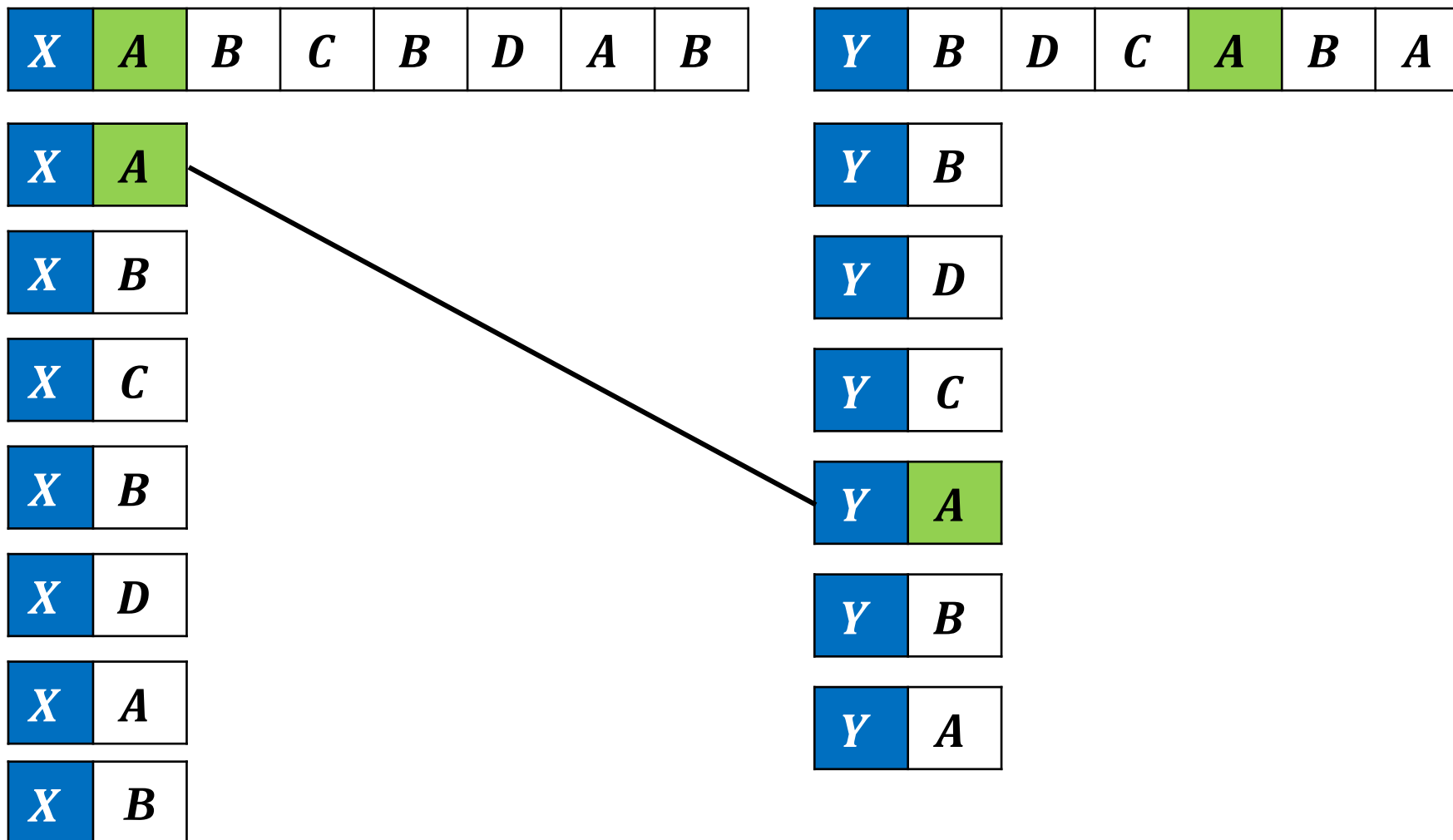
<i>X</i>	<i>A</i>
----------	----------

<i>Y</i>	<i>A</i>
----------	----------

<i>X</i>	<i>B</i>
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枚举并检查长度为1的子序列

- 枚举所有子序列



枚举并检查长度为1的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i>
----------	----------	----------	----------	----------

...

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

...

<i>X</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>
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<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>B</i>
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...

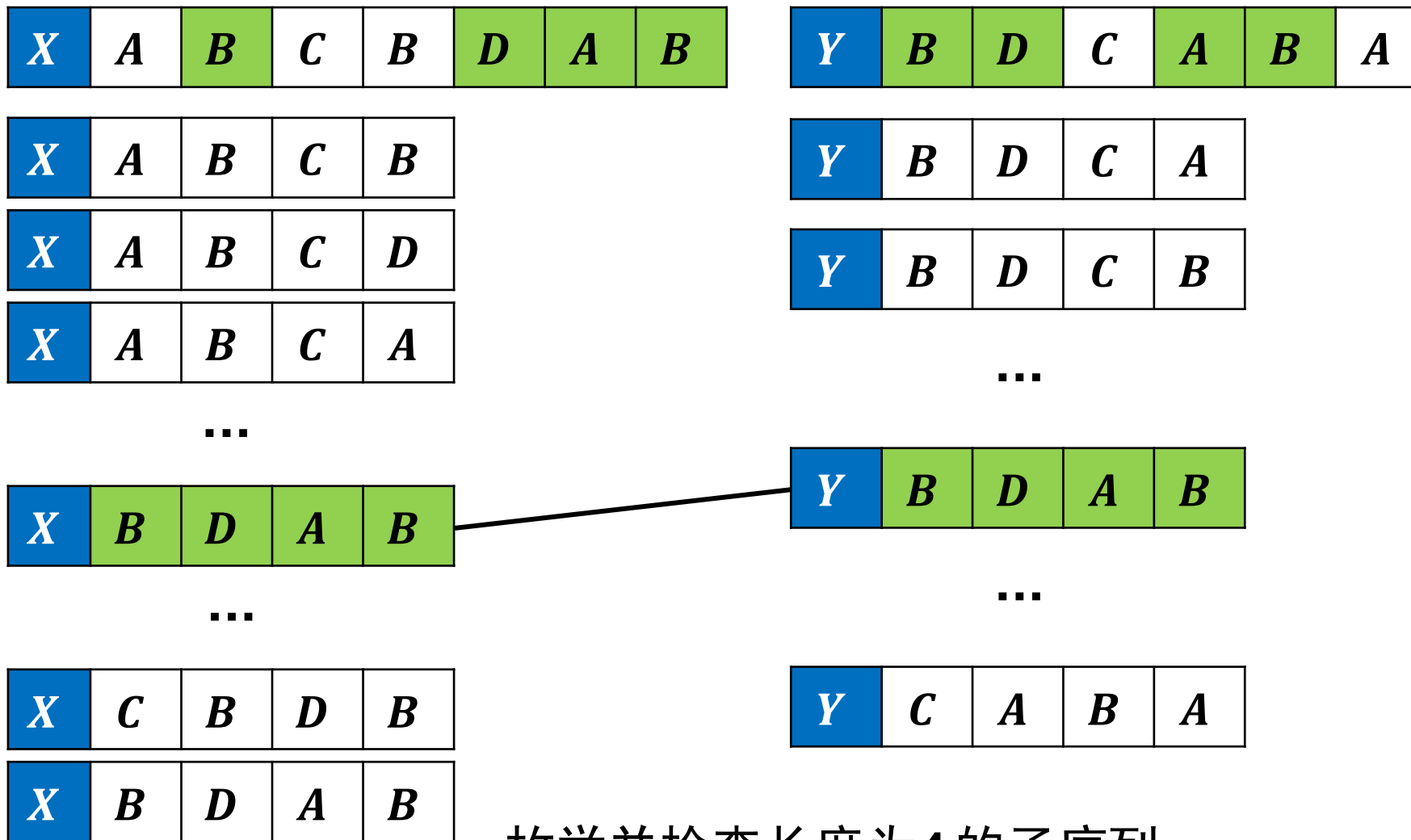
<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
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...

<i>Y</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
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枚举并检查长度为4的子序列

- 枚举所有子序列



枚举并检查长度为4的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i>
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...

...

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
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...

...

<i>X</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
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枚举并检查长度为4的子序列

- 枚举所有子序列

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

X	A	B	C	B	D
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X	A	B	C	B	A
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X	A	B	C	B	B
----------	---	---	---	---	---

X	A	B	B	D	A
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X	A	B	B	D	B
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...

X	C	B	D	A	B
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Y	B	D	C	A	B	A
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Y	B	D	C	A	B
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Y	B	D	C	A	A
----------	---	---	---	---	---

Y	B	D	C	B	A
----------	---	---	---	---	---

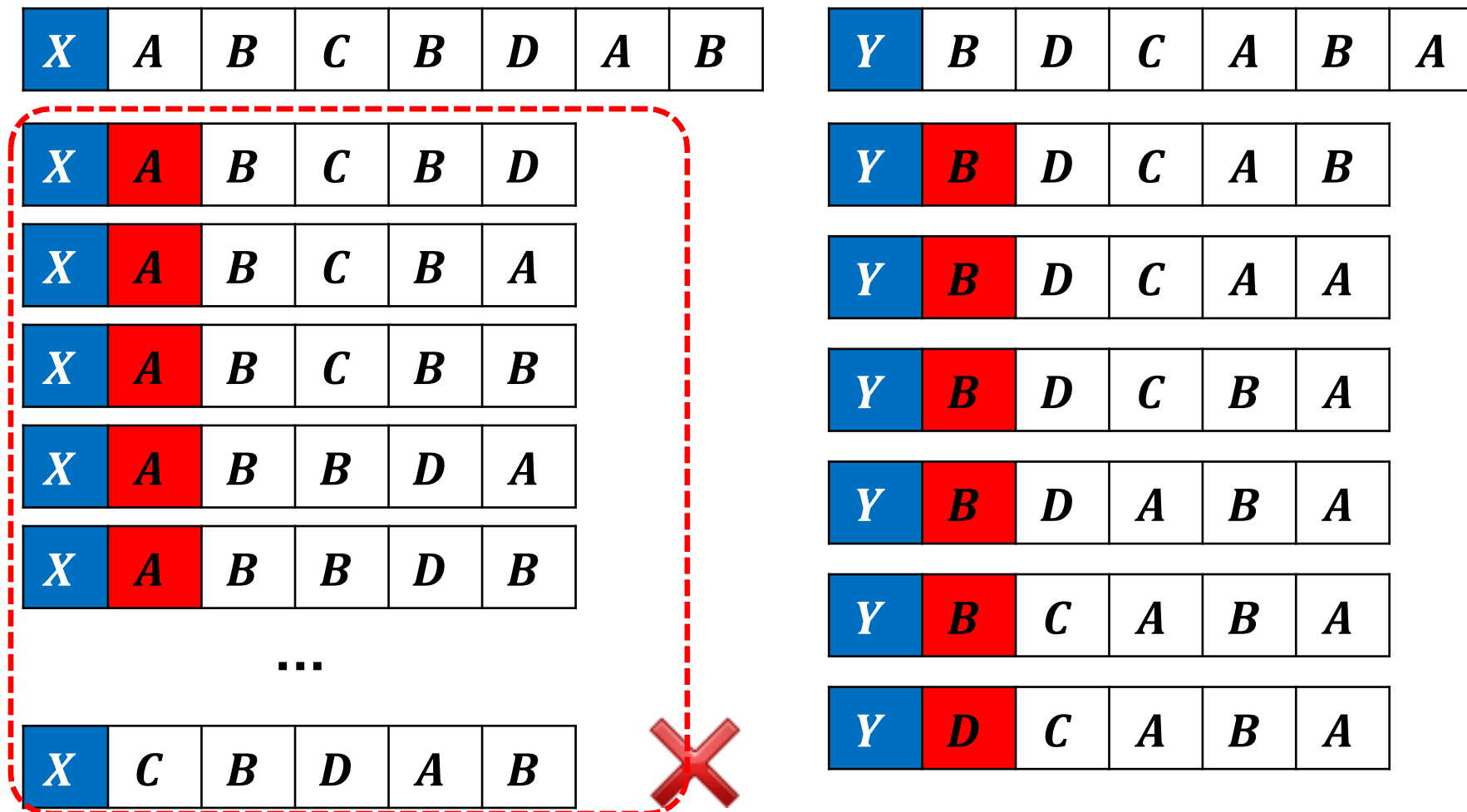
Y	B	D	A	B	A
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Y	B	C	A	B	A
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Y	D	C	A	B	A
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枚举并检查长度为5的子序列

- 枚举所有子序列



枚举并检查长度为5的子序列

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>B</i>
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<i>Y</i>	<i>B</i>
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长度为1

<i>X</i>	<i>A</i>	<i>B</i>
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<i>Y</i>	<i>A</i>	<i>B</i>
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长度为2

<i>X</i>	<i>A</i>	<i>B</i>	<i>A</i>
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<i>Y</i>	<i>A</i>	<i>B</i>	<i>A</i>
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长度为3

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
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长度为4

<i>X</i>					
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<i>Y</i>					
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长度为5

<i>X</i>						
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<i>Y</i>						
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长度为6



- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>B</i>
----------	----------

<i>X</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
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长度为1

长度为2

长度为3

长度为4

- 枚举所有子序列

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>B</i>
----------	----------

长度为1

<i>X</i>	<i>A</i>	<i>B</i>
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<i>Y</i>	<i>A</i>	<i>B</i>
----------	----------	----------

长度为2

<i>X</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>	<i>A</i>
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长度为3

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

长度为4

最长公共子序列



<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

长度为4

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

长度为4

长度为3

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>
----------	----------	----------

长度为4

长度为3

长度为2

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>Y</i>	<i>B</i>
----------	----------

长度为4

长度为3

长度为2

长度为1

<i>X</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>X</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>X</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>X</i>	<i>B</i>
----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------

<i>Y</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------

<i>Y</i>	<i>A</i>	<i>B</i>
----------	----------	----------

<i>Y</i>	<i>B</i>
----------	----------

长度为4

长度为3

长度为2

长度为1

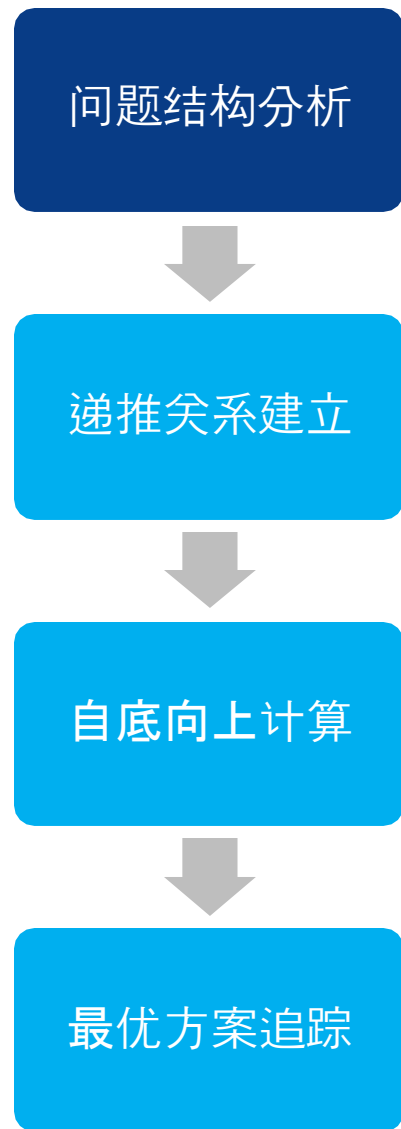
- 可能存在最优子结构和重叠子问题

问题：如何利用动态规划求解？

- 给出问题表示
 - $C[i, j]$: $X[1..i]$ 和 $Y[1..j]$ 的最长公共子序列长度

X	x_1	x_2	$\dots$	x_{i-1}	x_i
Y	y_1	y_2	$\dots$	y_{j-1}	y_j

- 明确原始问题
 - $C[n, m]$: $X[1..n]$ 和 $Y[1..m]$ 的最长公共子序列长度



递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况1: $x_7 \neq y_6$

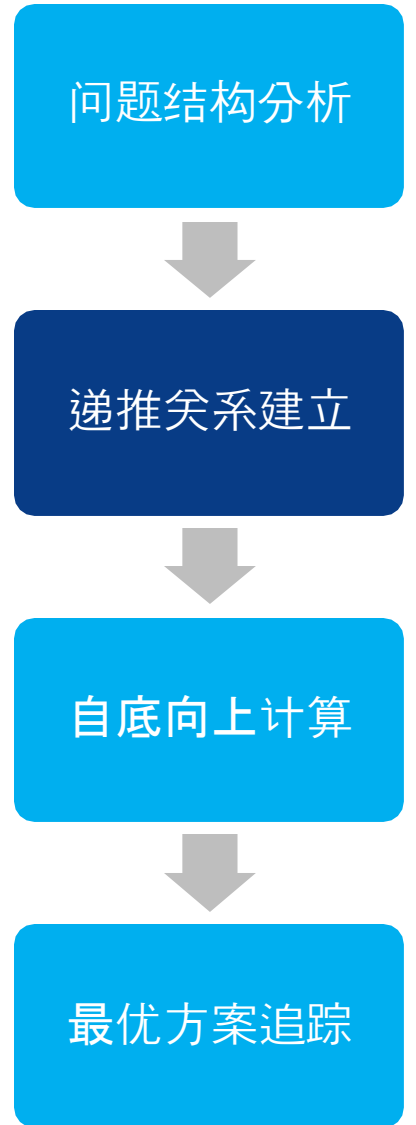
<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

- 情况2: $x_7 = y_6$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------



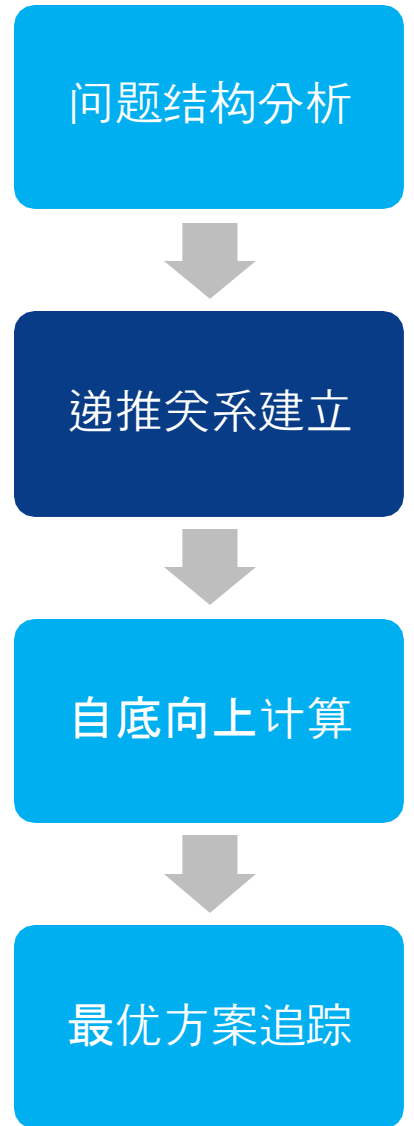
递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况1: $x_7 \neq y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
	Y	B	D	C	A	B	A

不同时出现



递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况1: $x_7 \neq y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

Y	B	D	C	A	B	A
----------	---	---	---	---	---	----------

$C[7, 6 - 1] + 0$							
X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况1: $x_7 \neq y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

Y	B	D	C	A	B	A
----------	---	---	---	---	---	----------

$C[7, 6 - 1] + 0$

X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

$C[7 - 1, 6] + 0$

X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况1: $x_7 \neq y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

max

Y	B	D	C	A	B	A
----------	---	---	---	---	---	----------

$C[7, 6 - 1] + 0$

X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

$C[7 - 1, 6] + 0$

X	A	B	C	B	D	A	B
Y	B	D	C	A	B	A	

问题结构分析

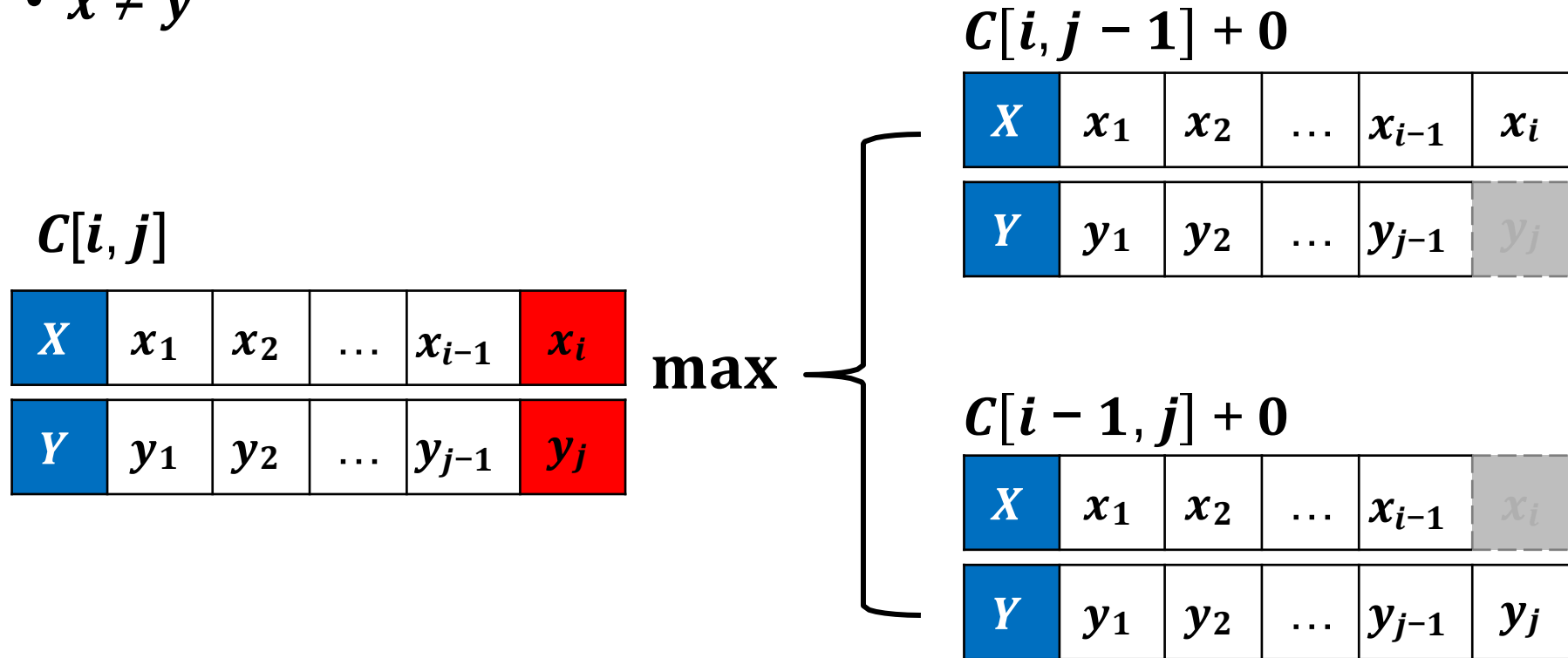
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

• $x \neq y$



$$C[i, j] = \max\{C[i - 1, j], C[i, j - 1]\}$$

最优子结构

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况2: $x_7 = y_6$

$C[7, 6]$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
-----------------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>
-----------------	----------	----------	----------	----------	----------	----------



递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况2: $x_7 = y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
	Y	B	D	C	A	B	B

可同时出现



递推关系建立：分析最优（子）结构

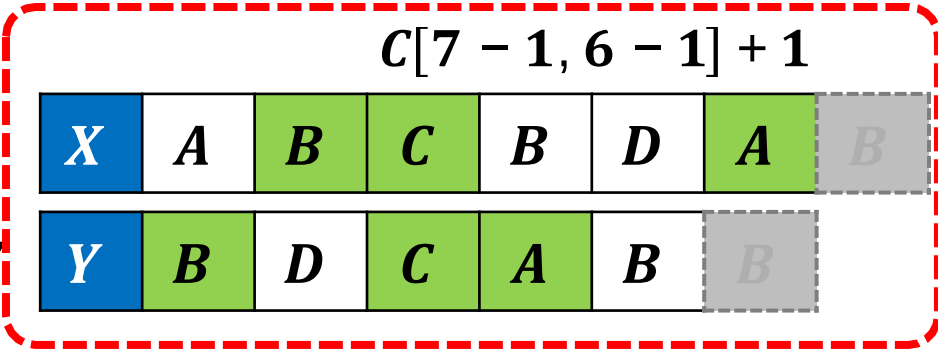
- 考察末尾字符
 - 情况2: $x_7 = y_6$

$C[7, 6]$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------

可同时出现



递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况2: $x_7 = y_6$

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

$C[7, 6]$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------

也可不同时出现

$C[7 - 1, 6 - 1] + 1$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>	

$C[7, 6 - 1] + 0$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>	

$C[7 - 1, 6] + 0$

<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
<i>Y</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>B</i>	

递推关系建立：分析最优（子）结构

- 考察末尾字符
 - 情况2: $x_7 = y_6$

$C[7, 6]$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

Y	B	D	C	A	B	B
----------	---	---	---	---	---	----------

max

$$C[7 - 1, 6 - 1] + 1$$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

Y	B	D	C	A	B	B
----------	---	---	---	---	---	---

$$C[7, 6 - 1] + 0$$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

Y	B	D	C	A	B	B
----------	---	---	---	---	---	---

$$C[7 - 1, 6] + 0$$

X	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

Y	B	D	C	A	B	B
----------	---	---	---	---	---	---

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- $x = y$

$$C[i, j]$$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

max

$$C[i-1, j-1] + 1$$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

$$C[i-1, j] + 0$$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

$$C[i, j-1] + 0$$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

思考：3个问题是否都需要求解？

递推关系建立：分析最优（子）结构

- $x = y$
 - $C[i-1, j]$ 比 $C[i-1, j-1]$ 至多大1
 - $C[i, j-1]$ 比 $C[i-1, j-1]$ 至多大1

$C[i, j]$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

max

$C[i-1, j-1] + 1$					
X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j
$C[i-1, j] + 0$					
X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j
$C[i, j-1] + 0$					
X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- $x = y$
 - $C[i-1, j]$ 比 $C[i-1, j-1]$ 至多大 1
 - $C[i, j-1]$ 比 $C[i-1, j-1]$ 至多大 1
 - $C[i-1, j-1] + 1$, 另外两个 +0

$C[i, j]$

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

max

$C[i-1, j-1] + 1$ 已充分

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

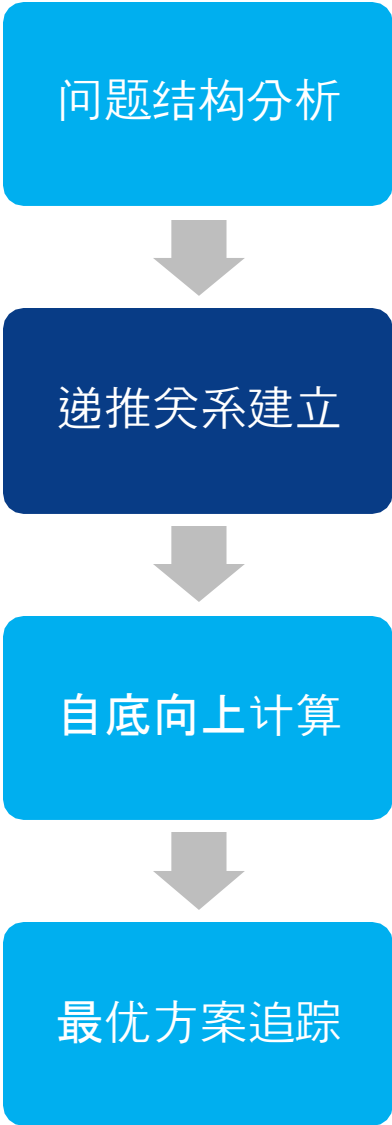
$C[i-1, j] + 0$ 非必要

X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

$C[i, j-1] + 0$ 非必要

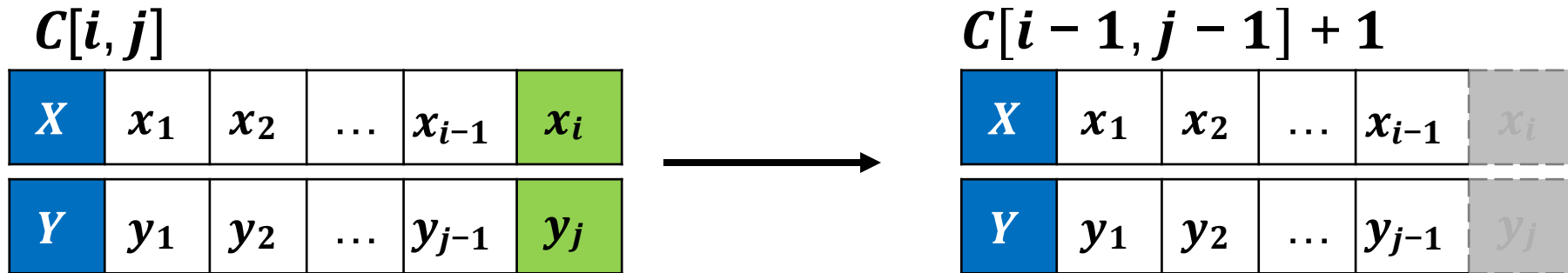
X	x_1	x_2	...	x_{i-1}	x_i
Y	y_1	y_2	...	y_{j-1}	y_j

$C[i-1, j-1] + 1$
 $\geq \max\{C[i, j-1], C[i-1, j]\}$



递推关系建立：分析最优（子）结构

- $x = y$



$$C[i, j] = C[i-1, j-1] + 1$$

最优子结构

问题结构分析

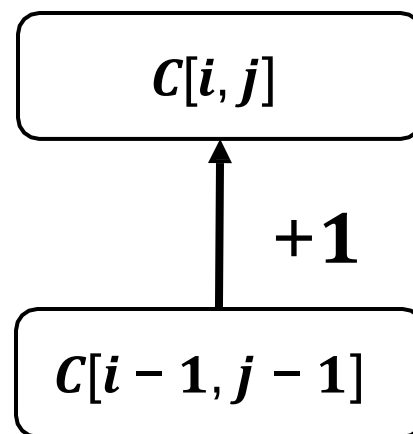
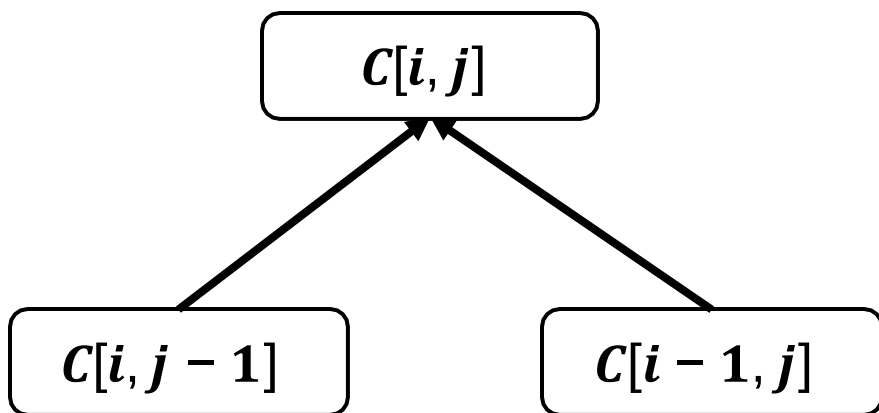
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：构造递推公式

$$\bullet C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$



问题结构分析

递推关系建立

自底向上计算

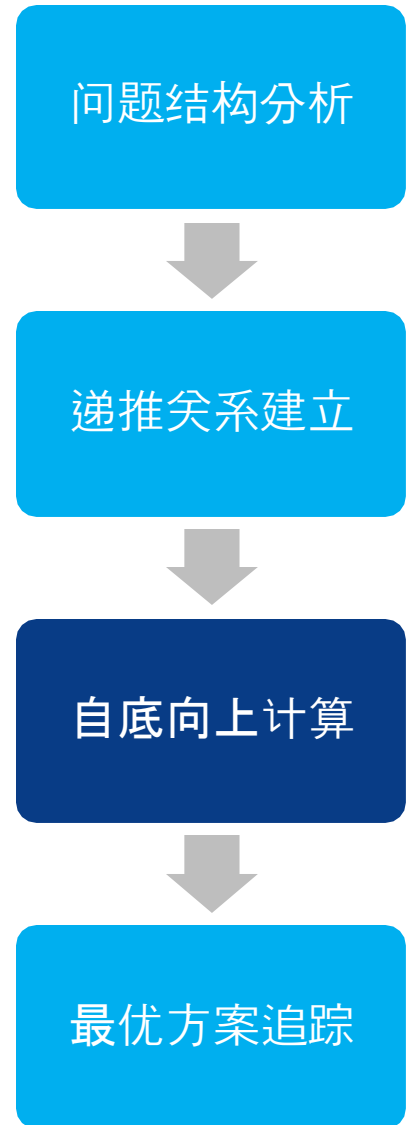
最优方案追踪

自底向上计算：确定计算顺序

- 初始化

- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$					
...					
$i = n$					



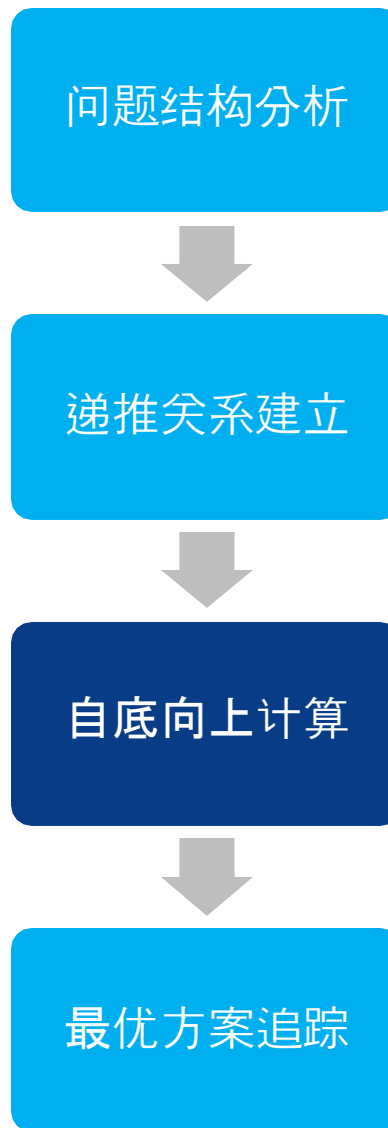
自底向上计算：确定计算顺序

• 初始化

- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0				
$i = 2$	0				
...	0				
$i = n$	0				

初始化



自底向上计算：确定计算顺序

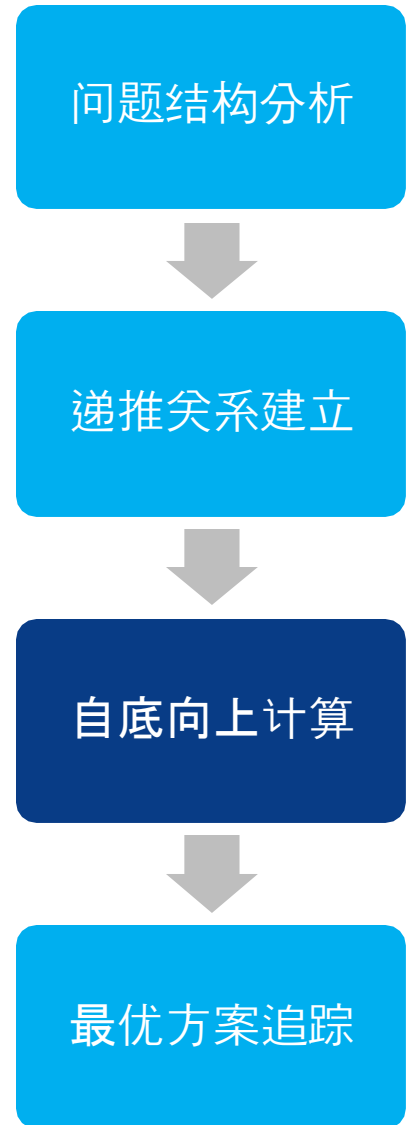
- 初始化

- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

- 递推公式

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0				
$i = 2$	0				
...	0				
$i = n$	0				



自底向上计算：确定计算顺序

- 初始化

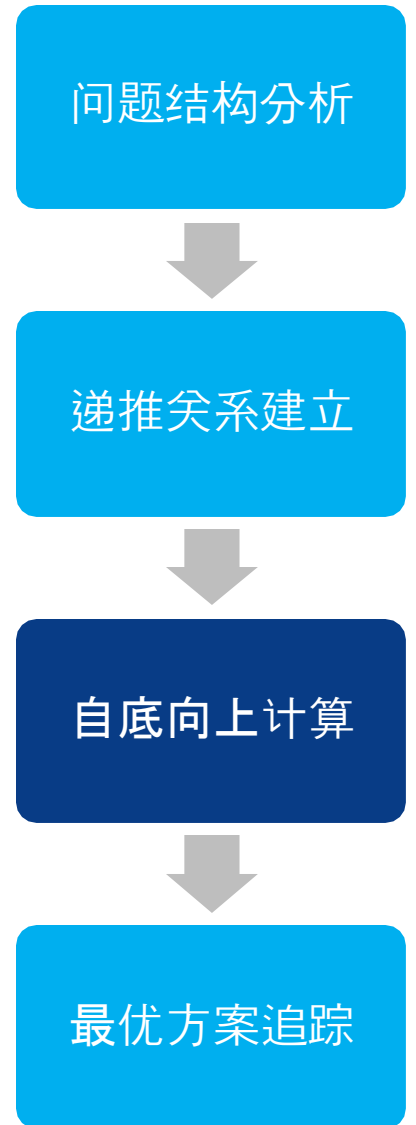
- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

- 递推公式

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0				
$i = 2$	0				
...	0				
$i = n$	0				

Diagram illustrating the dynamic programming table structure. A black arrow points to the cell $C[i, j]$ (row i , column j), and a red arrow points to the cell $C[i-1, j-1]$ (row $i-1$, column $j-1$).



自底向上计算：确定计算顺序

- 初始化

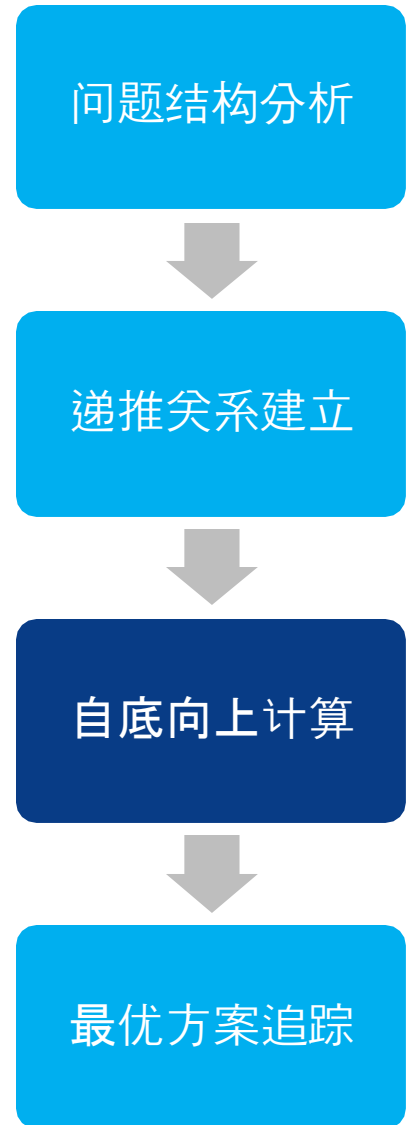
- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

- 递推公式

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0				
$i = 2$	0				
...	0				
$i = n$	0				

Diagram illustrating the calculation order for $C[i, j]$. A red arrow points from the left, and a black arrow points from above, both converging on the cell $C[i, j]$ in the row $i=2$ and column $j=2$.



自底向上计算：确定计算顺序

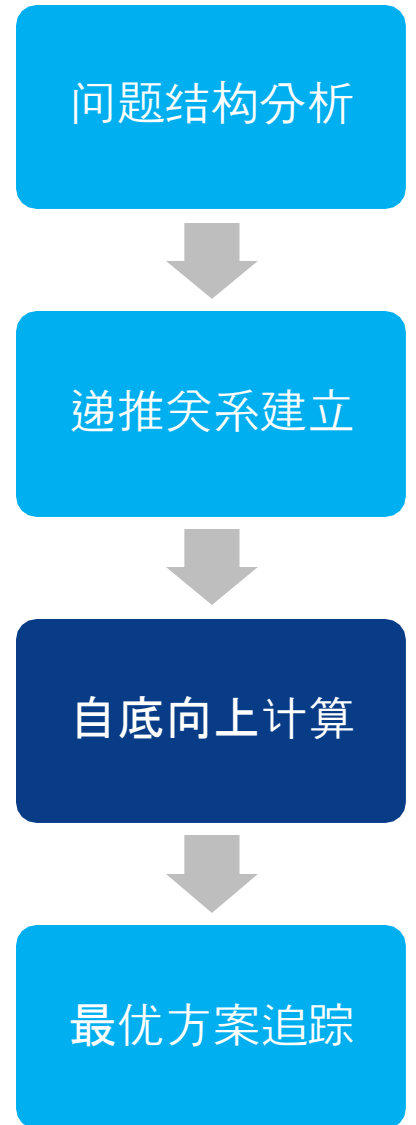
- 初始化

- $C[0, i] = C[0, j] = 0$
- 某序列长度为0时，最长公共子序列长度为0

- 递推公式

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0				
$i = 2$	0				
...	0				
$i = n$	0				

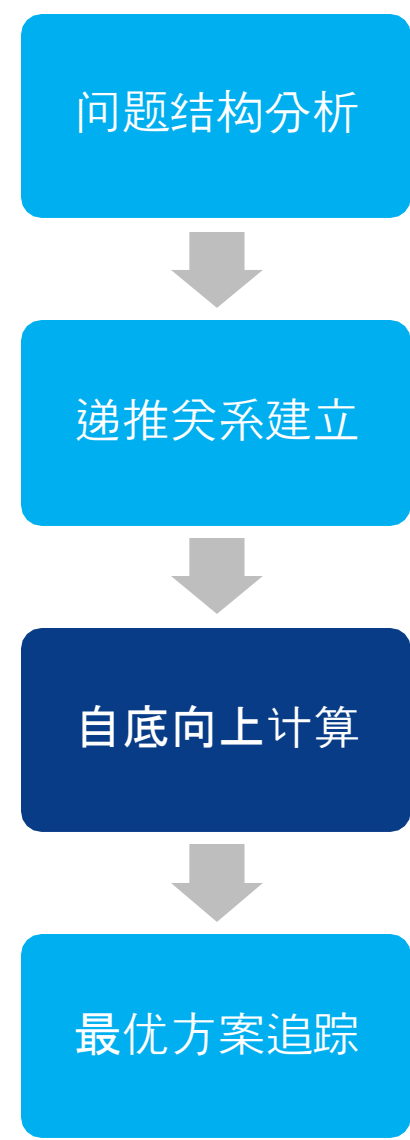


自底向上计算：依次求解问题

- 初始化
 - $C[0, i] = C[0, j] = 0$
 - 某序列长度为0时，最长公共子序列长度为0
- 递推公式
 - $$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$	0	0	0	0	0
$i = 1$	0	→			
$i = 2$	0	←	→	→	→
...	0	←	→	→	→
$i = n$	0	←	→	→	→ ★

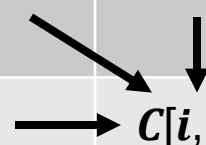
自底向上计算

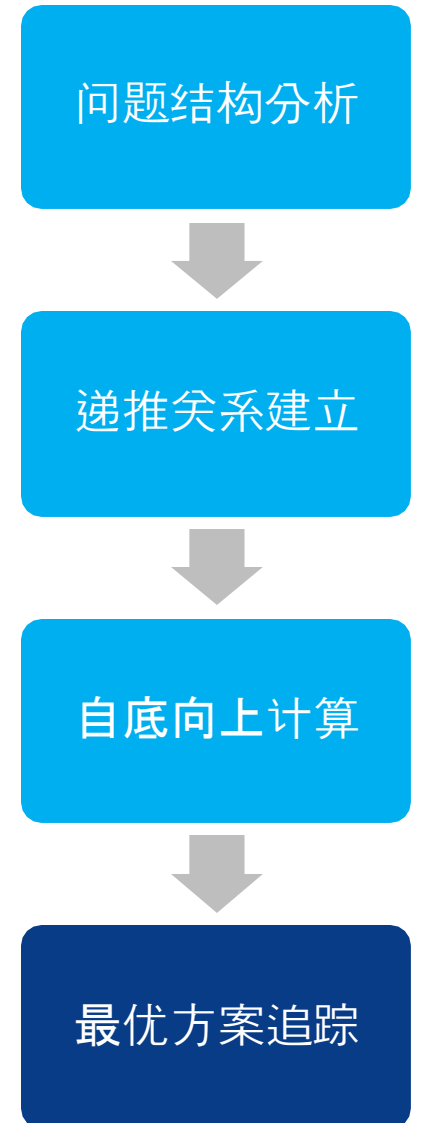


最优方案追踪：记录决策过程

- 构造追踪数组 $rec[1..n]$ ，记录子问题来源
- $$rec[i, j] = \begin{cases} LU, & \text{if } C[i, j] = C[i - 1, j - 1] + 1 \\ U, & \text{if } C[i, j] = C[i - 1, j] \\ L, & \text{if } C[i, j] = C[i, j - 1] \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$					
...					
$i = n$					





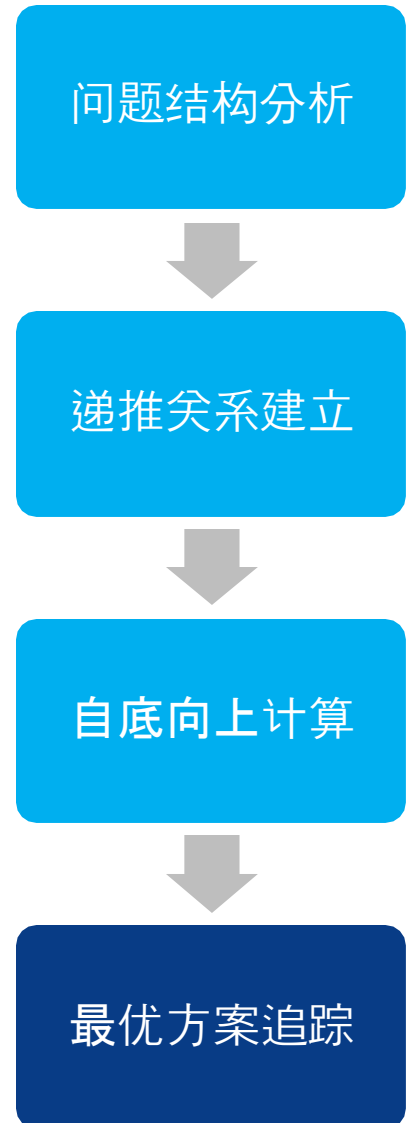
最优方案追踪：记录决策过程

- 构造追踪数组 $rec[1..n]$ ，记录子问题来源
- $$rec[i, j] = \begin{cases} LU, & \text{if } C[i, j] = C[i-1, j-1] + 1 \\ U, & \text{if } C[i, j] = C[i-1, j] \\ L, & \text{if } C[i, j] = C[i, j-1] \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$					
...					
$i = n$					



最长公共子序列末尾为 $X[i] = Y[j]$



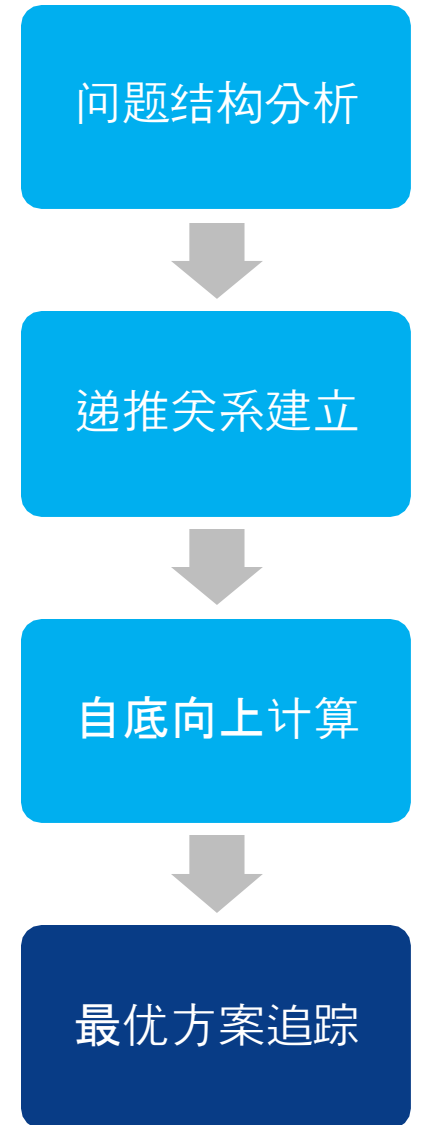
最优方案追踪：记录决策过程

- 构造追踪数组 $rec[1..n]$ ，记录子问题来源
- $$rec[i, j] = \begin{cases} LU, & \text{if } C[i, j] = C[i - 1, j - 1] + 1 \\ U, & \text{if } C[i, j] = C[i - 1, j] \\ L, & \text{if } C[i, j] = C[i, j - 1] \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$					
...					
$i = n$					



最长公共子序列在 $X[1..i - 1]$ 和 $Y[1..j]$ 中

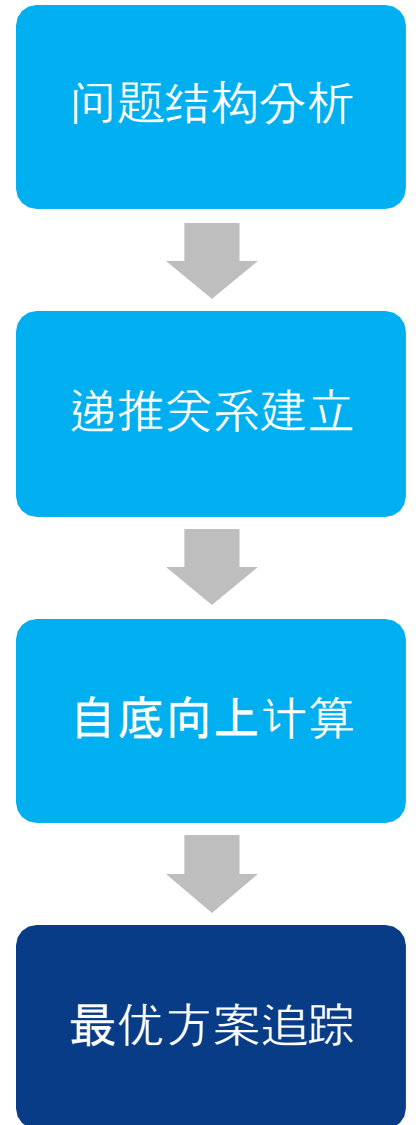


最优方案追踪：记录决策过程

- 构造追踪数组 $rec[1..n]$ ，记录子问题来源
- $$rec[i, j] = \begin{cases} LU, & \text{if } C[i, j] = C[i - 1, j - 1] + 1 \\ U, & \text{if } C[i, j] = C[i - 1, j] \\ L, & \text{if } C[i, j] = C[i, j - 1] \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$					
...					
$i = n$					

最长公共子序列在 $X[1..i]$ 和 $Y[1..j-1]$ 中

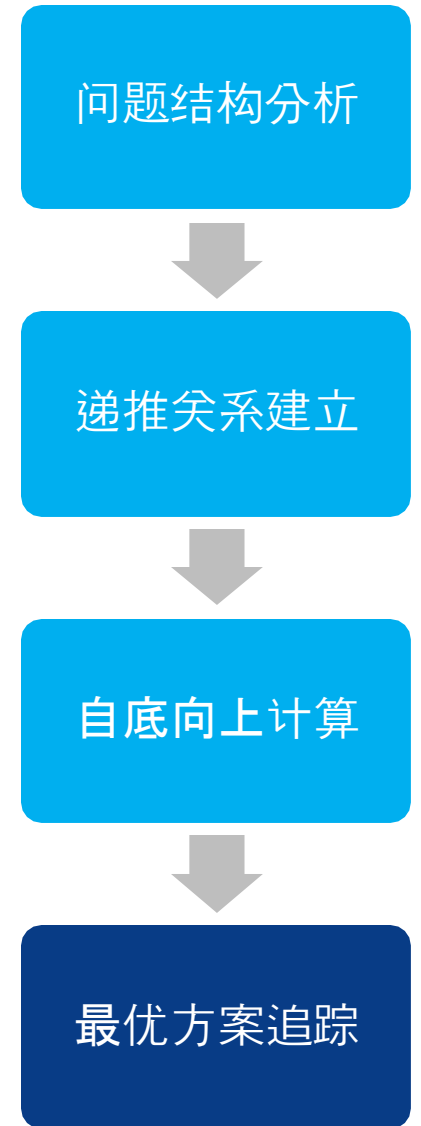


最优方案追踪：记录决策过程

- 构造追踪数组 $rec[1..n]$ ，记录子问题来源
- $$rec[i, j] = \begin{cases} LU, & \text{if } C[i, j] = C[i - 1, j - 1] + 1 \\ U, & \text{if } C[i, j] = C[i - 1, j] \\ L, & \text{if } C[i, j] = C[i, j - 1] \end{cases}$$

$C[i, j]$	$j = 0$	$j = 1$	$j = 2$	...	$j = m$
$i = 0$					
$i = 1$					
$i = 2$				$rec[] = LU$	
...					
$i = n$			$rec[] = U$		

$rec[] = L$



	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0							
1							
2							
3							
4							
5							
6							
7							

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
Y_j	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0						
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

初始化

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$X_i \neq Y_j$

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0						
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$X_i \neq Y_j$

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0						
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$X_i \neq Y_j$

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0						
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$C[1, 1] = \max\{C[1, 0], C[0, 1]\}$

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U					
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$X_i \neq Y_j$

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0					
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U					
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0				
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U				
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0			
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U			
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$X_i = Y_j$ $rec[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1		
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$$C[1, 4] = C[0, 3] + 1$$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU		
2						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1		
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0						
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2						
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1					
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU					
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1				
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L				
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1			
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L			
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1		
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U		
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0						
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3						
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1					
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U					
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1				
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U				
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2			
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU			
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2		
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L		
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0						
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4						
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1					
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU					
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1				
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U				
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2			
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U			
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2		
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U		
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0						
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5						
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1					
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U					
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2				
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU				
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2			
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U			
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2		
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U		
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0						
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6						
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1					
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U					
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2				
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U				
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2			
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U			
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3		
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU		
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0						

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7						

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1					

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU					

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2				

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U				

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2			

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U			

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3		

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U		

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	U	U	U	U	LU	U

最长公共子序列的长度

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

--	--	--	--

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

			A
--	--	--	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

			A
--	--	--	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

		B	A
--	--	---	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

		B	A
--	--	---	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

	C	B	A
--	---	---	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

	C	B	A
--	---	---	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

B	C	B	A
---	---	---	---

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

	1	2	3	4	5	6	7
X_i	A	B	C	B	D	A	B
Y_j	B	D	C	A	B	A	

B	C	B	A
---	---	---	---

最长公共子序列

$C[]$

$j \backslash i$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	1
2	0	1	1	1	1	2	2
3	0	1	1	2	2	2	2
4	0	1	1	2	2	3	3
5	0	1	2	2	2	3	3
6	0	1	2	2	3	3	4
7	0	1	2	2	3	4	4

$rec[]$

$j \backslash i$	1	2	3	4	5	6
1	U	U	U	LU	L	LU
2	LU	L	L	U	LU	L
3	U	U	LU	L	U	U
4	LU	U	U	U	LU	L
5	U	LU	U	U	U	U
6	U	U	U	LU	U	LU
7	LU	U	U	U	LU	U

- Longest-Common-Subsequence(X, Y)

输入: 两个序列 X, Y

输出: X 和 Y 的最长公共子序列

$n \leftarrow \text{length}(X)$

$m \leftarrow \text{length}(Y)$

//初始化

新建二维数组 $C[0..n, 0..m]$ 和 $rec[0..n, 0..m]$

for $i \leftarrow 0$ to n do

$C[i, 0] \leftarrow 0$

end

for $j \leftarrow 0$ to m do

$C[0, j] \leftarrow 0$

end

序列长度

- Longest-Common-Subsequence(X, Y)

输入: 两个序列 X, Y

输出: X 和 Y 的最长公共子序列

$n \leftarrow \text{length}(X)$

$m \leftarrow \text{length}(Y)$

//初始化

新建二维数组 $C[0..n, 0..m]$ 和 $rec[0..n, 0..m]$

for $i \leftarrow 0$ to n do

 | $C[i, 0] \leftarrow 0$

end

for $j \leftarrow 0$ to m do

 | $C[0, j] \leftarrow 0$

end

初始化

- Longest-Common-Subsequence(X, Y)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
    if  $X_i = Y_j$  then
       $C[i, j] \leftarrow C[i - 1, j - 1] + 1$ 
       $rec[i, j] \leftarrow "LU"$ 
    end
    else if  $C[i - 1, j] \geq C[i, j - 1]$  then
       $C[i, j] \leftarrow C[i - 1, j]$ 
       $rec[i, j] \leftarrow "U"$ 
    end
    else
       $C[i, j] \leftarrow C[i, j - 1]$ 
       $rec[i, j] \leftarrow "L"$ 
    end
  end
end
return  $C, rec$ 
```

依次计算子问题

- Longest-Common-Subsequence(X, Y)

//动态规划

for $i \leftarrow 1$ to n do

for $j \leftarrow 1$ to m do

if $X_i = Y_j$ then

$C[i, j] \leftarrow C[i - 1, j - 1] + 1$

$rec[i, j] \leftarrow "LU"$

end

else if $C[i - 1, j] \geq C[i, j - 1]$ then

$C[i, j] \leftarrow C[i - 1, j]$

$rec[i, j] \leftarrow "U"$

end

else

$C[i, j] \leftarrow C[i, j - 1]$

$rec[i, j] \leftarrow "L"$

end

end

end

return C, rec

末尾相等

- Longest-Common-Subsequence(X, Y)

//动态规划

for $i \leftarrow 1$ to n do

 for $j \leftarrow 1$ to m do

 if $X_i = Y_j$ then

$C[i, j] \leftarrow C[i - 1, j - 1] + 1$

$rec[i, j] \leftarrow "LU"$

 end

 else if $C[i - 1, j] \geq C[i, j - 1]$ then

$C[i, j] \leftarrow C[i - 1, j]$

$rec[i, j] \leftarrow "U"$

 end

 else

$C[i, j] \leftarrow C[i, j - 1]$

$rec[i, j] \leftarrow "L"$

 end

 end

end

return C, rec

记录长度和决策

- Longest-Common-Subsequence(X, Y)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
    if  $X_i = Y_j$  then
       $C[i, j] \leftarrow C[i - 1, j - 1] + 1$ 
       $rec[i, j] \leftarrow "LU"$ 
    end
    else if  $C[i - 1, j] \geq C[i, j - 1]$  then
       $C[i, j] \leftarrow C[i - 1, j]$ 
       $rec[i, j] \leftarrow "U"$ 
    end
    else
       $C[i, j] \leftarrow C[i, j - 1]$ 
       $rec[i, j] \leftarrow "L"$ 
    end
  end
end
return  $C, rec$ 
```

末尾不等

- **Print-LCS(rec, X, i, j)**

输入: 追踪数组 rec , 序列 X , 当前位置 i 和 j

输出: $X[1..i]$ 和 $Y[1..j]$ 的最长公共子序列

if $i = 0$ or $j = 0$ then

 return NULL

end

if $rec[i, j] = \text{"LU"}$ then

 Print-LCS($rec, X, i - 1, j - 1$)

 print x_i

end

else if $rec[i, j] = \text{"U"}$ then

 Print-LCS($rec, X, i - 1, j$)

end

else

 Print-LCS($rec, X, i, j - 1$)

end

倒序追踪方案

- **Print-LCS(rec, X, i, j)**

输入: 追踪数组 rec , 序列 X , 当前位置 i 和 j

输出: $X[1..i]$ 和 $Y[1..j]$ 的最长公共子序列

if $i = 0$ or $j = 0$ then

| return NULL

end

if $rec[i, j] = \text{"LU"}$ then

| Print-LCS($rec, X, i - 1, j - 1$)

| print x_i

end

else if $rec[i, j] = \text{"U"}$ then

| Print-LCS($rec, X, i - 1, j$)

end

else

| Print-LCS($rec, X, i, j - 1$)

end

递归终止：序列长度为0

- **Print-LCS(*rec*, *X*, *i*, *j*)**

输入: 追踪数组 rec , 序列 X , 当前位置 i 和 j

输出: $X[1..i]$ 和 $Y[1..j]$ 的最长公共子序列

if $i = 0$ or $j = 0$ then

| return NULL

end

if $rec[i, j] = \text{"LU"}$ then

| Print-LCS($rec, X, i - 1, j - 1$)

| print x_i

end

else if $rec[i, j] = \text{"U"}$ then

| Print-LCS($rec, X, i - 1, j$)

end

else

| Print-LCS($rec, X, i, j - 1$)

end

追踪方案：末尾相等

- **Print-LCS(rec, X, i, j)**

输入: 追踪数组 rec , 序列 X , 当前位置 i 和 j

输出: $X[1..i]$ 和 $Y[1..j]$ 的最长公共子序列

if $i = 0$ or $j = 0$ then

| return NULL

end

if $rec[i, j] = \text{"LU"}$ then

| Print-LCS($rec, X, i - 1, j - 1$)

| print x_i

end

else if $rec[i, j] = \text{"U"}$ then

| Print-LCS($rec, X, i - 1, j$)

end

else

| Print-LCS($rec, X, i, j - 1$)

end

追踪方案：末尾不等

- Longest-Common-Subsequence(X, Y)

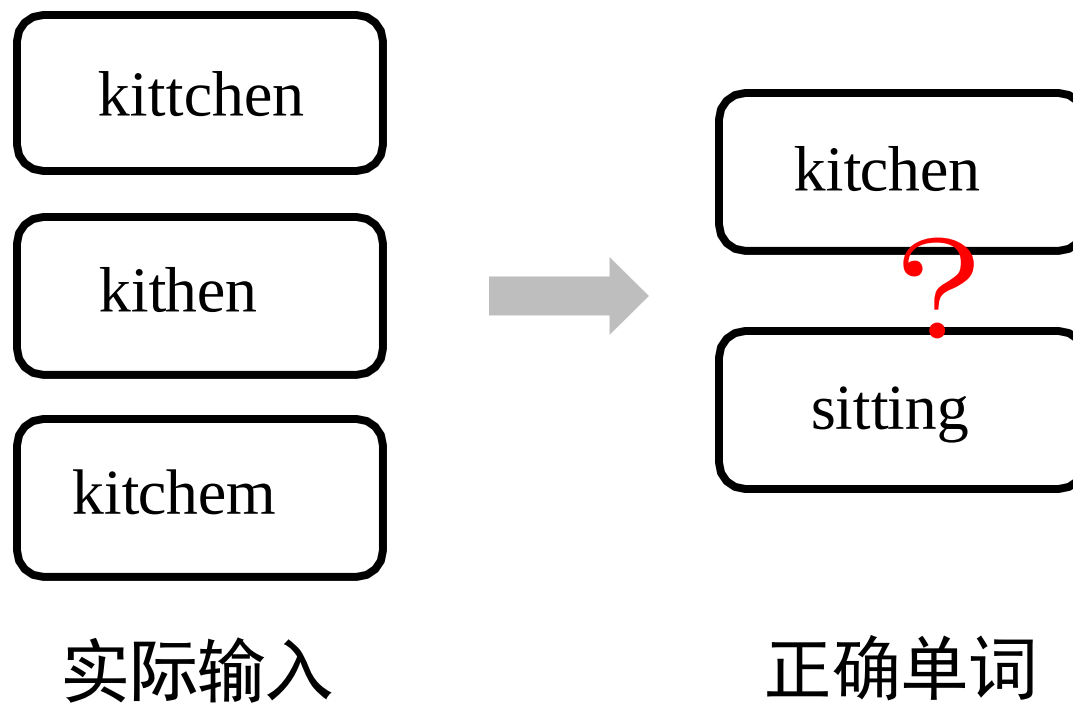
//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
    if  $X_i = Y_j$  then
       $C[i, j] \leftarrow C[i - 1, j - 1] + 1$ 
       $rec[i, j] \leftarrow "LU"$ 
    end
    else if  $C[i - 1, j] \geq C[i, j - 1]$  then
       $C[i, j] \leftarrow C[i - 1, j]$ 
       $rec[i, j] \leftarrow "U"$ 
    end
    else
       $C[i, j] \leftarrow C[i, j - 1]$ 
       $rec[i, j] \leftarrow "L"$ 
    end
  end
end
return  $C, rec$ 
```

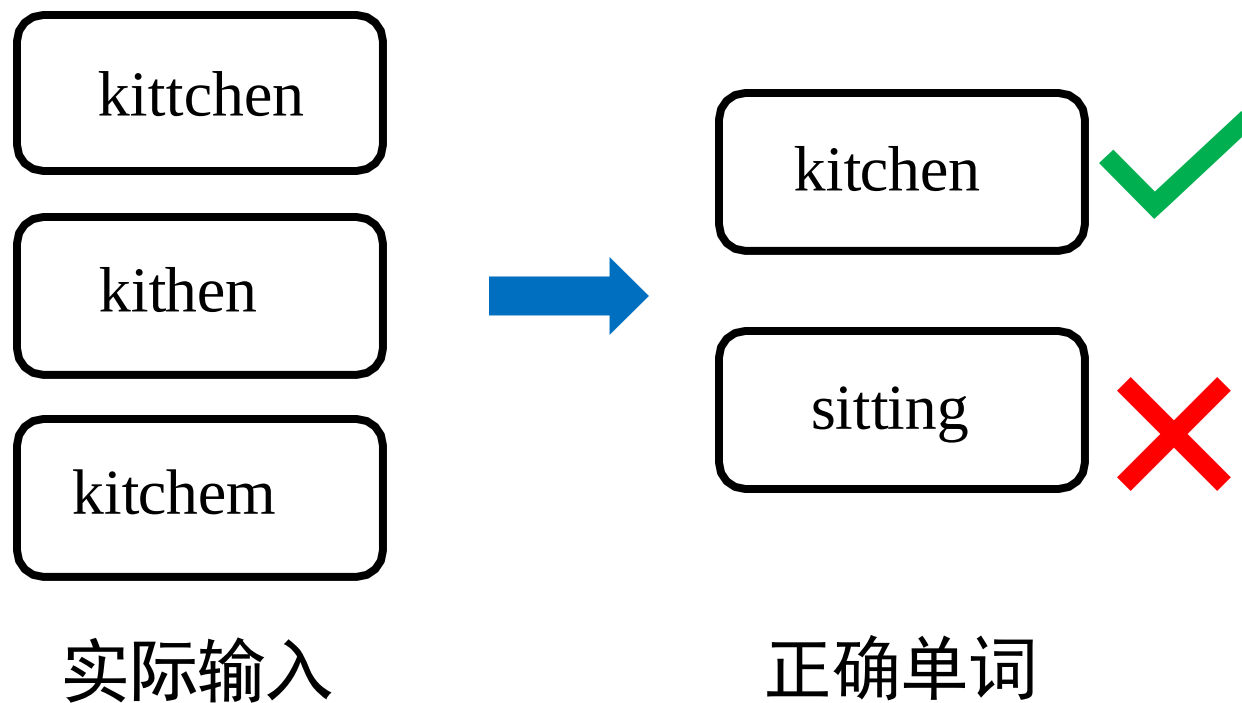
时间复杂度 : $O(n \cdot m)$

- 5.1 0-1背包问题
- 5.2 动态规划主要思想
- 5.3 再论背包问题
- 5.4 可靠性设计
- 5.5 最短路径问题
- 5.6 再论货郎担问题
- 5.7 最长公共子序列
- **5.8 编辑距离问题**
- 5.9 最优二分检索树
- 5.10 矩阵链乘积问题

- 输入法自动更正

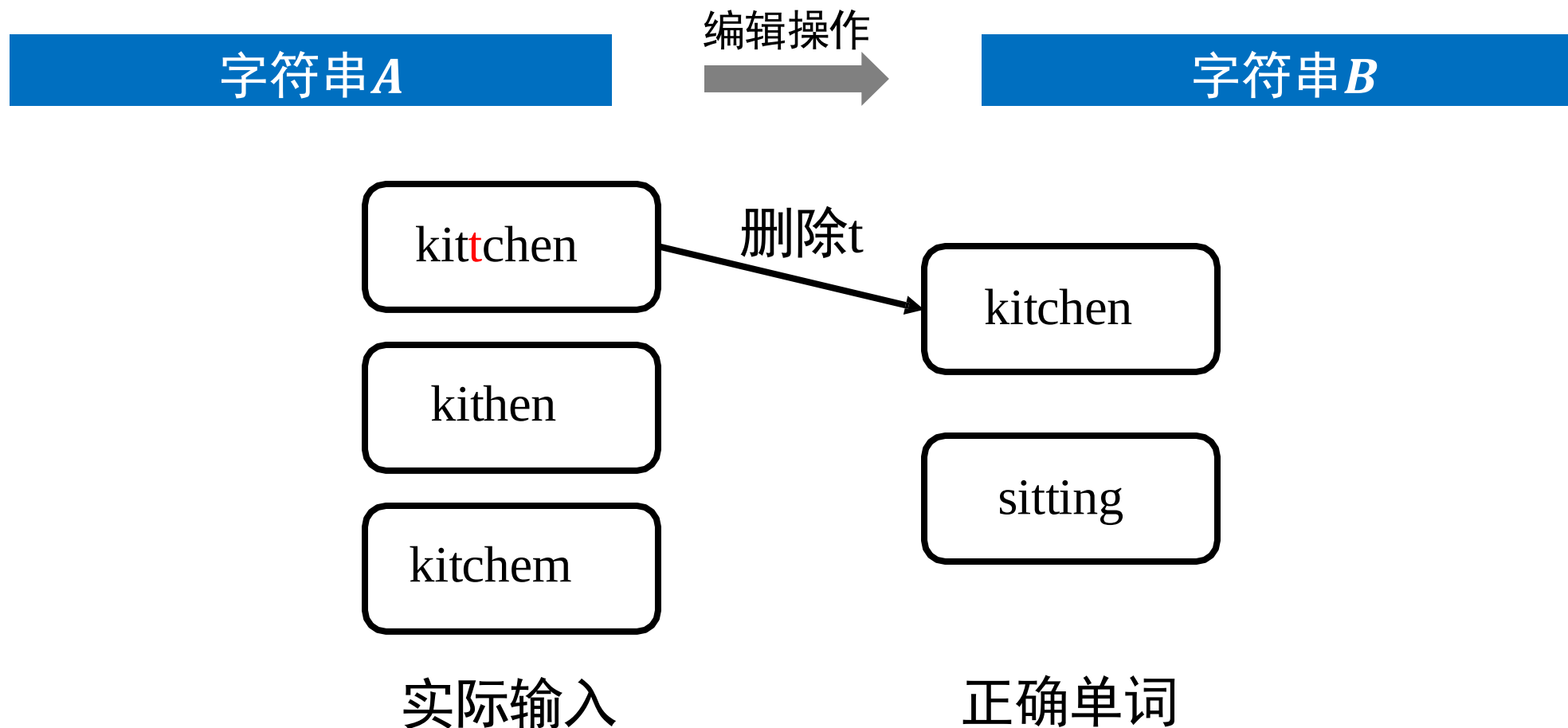


- 输入法自动更正



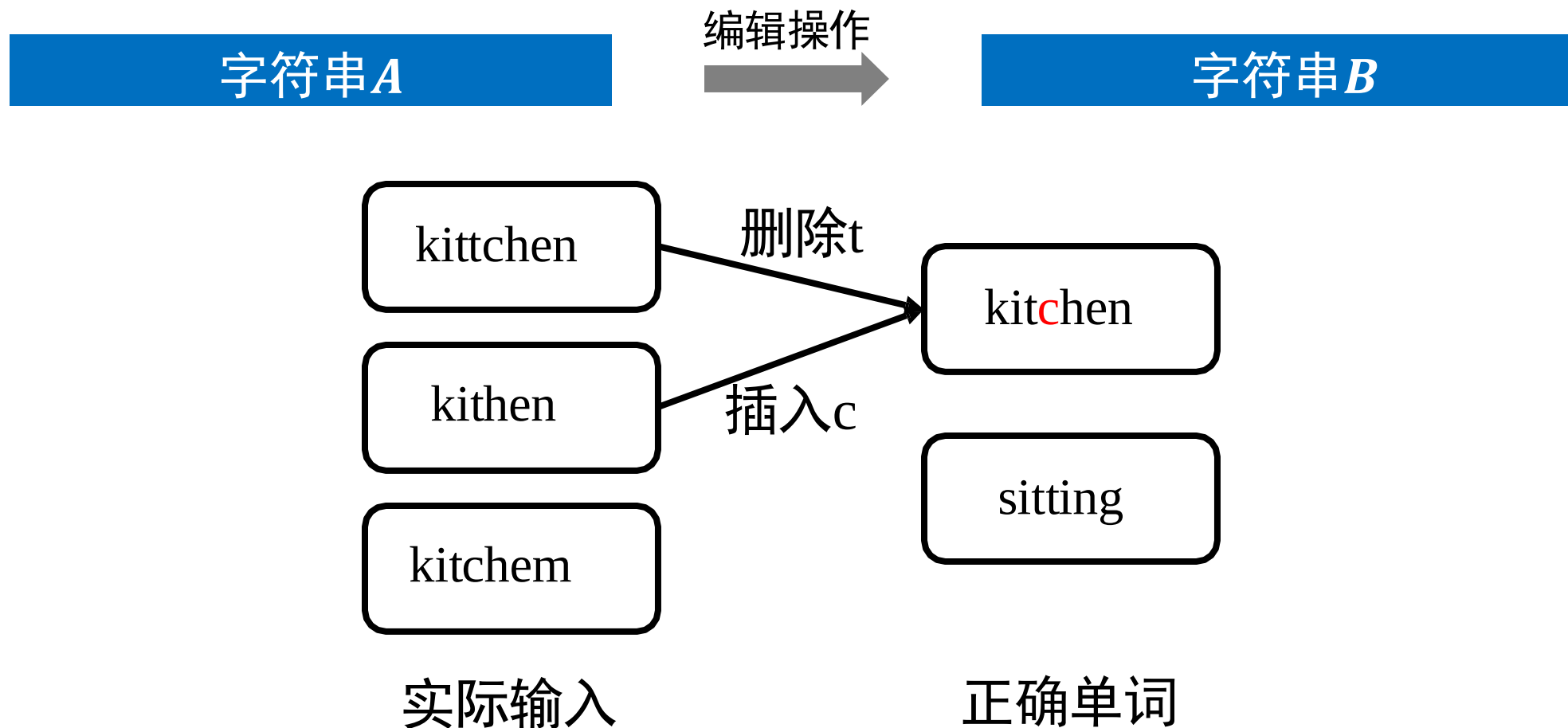
问题：如何衡量序列的相似程度？

- 基本思想



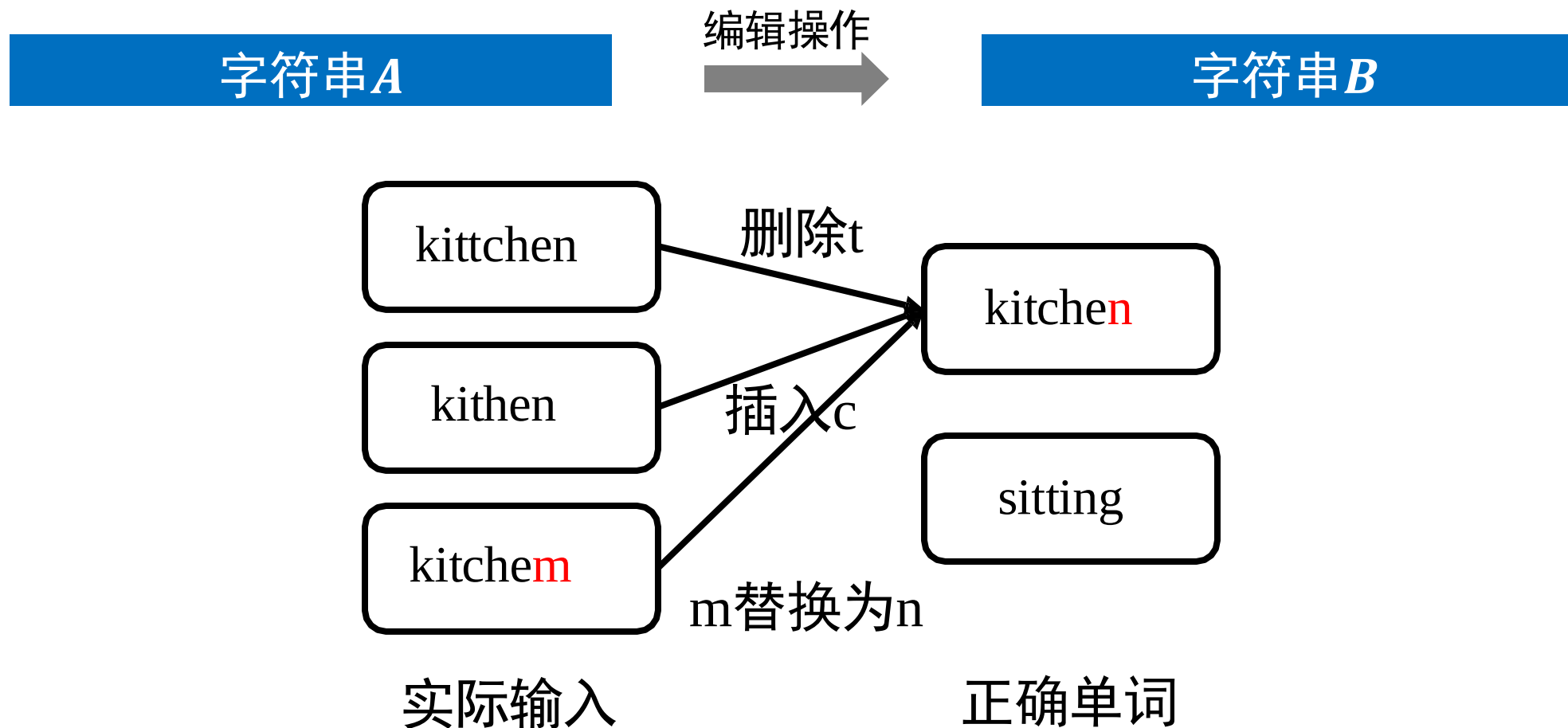
● 编辑操作：删除、插入、替换

- 基本思想



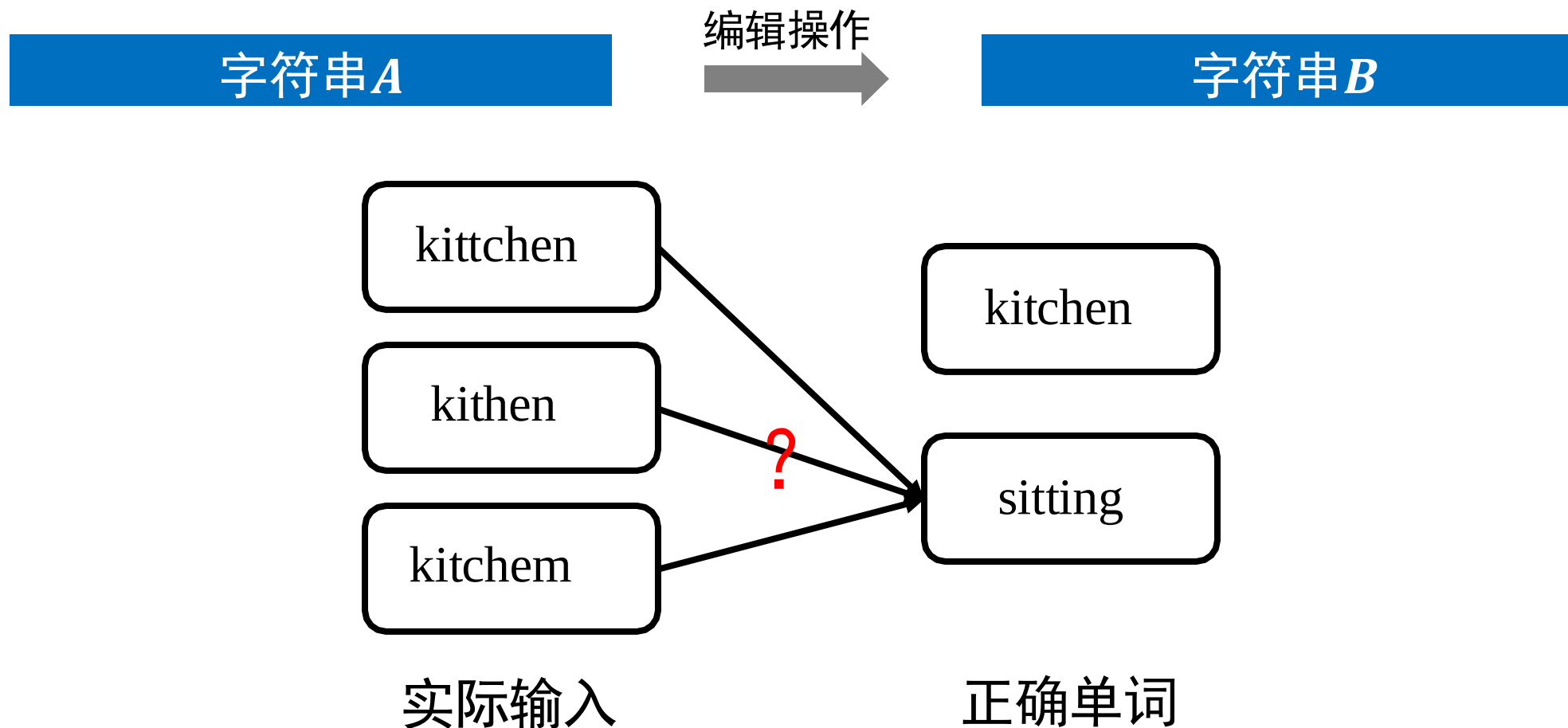
● 编辑操作：删除、插入、替换

- 基本思想



● 编辑操作：删除、插入、替换

- 基本思想



● 编辑操作：删除、插入、替换

编辑操作示例

$A = \text{kittchen}$

编辑操作

$B = \text{sitting}$

操作名称

操作示例

删除

kittchen $\rightarrow$ kitchen

插入

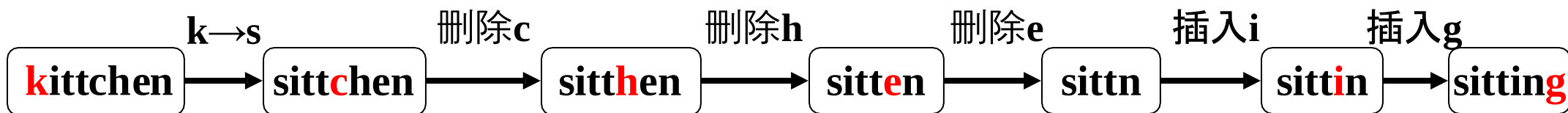
kithen $\rightarrow$ kit**ch**en

替换

kitchem $\rightarrow$ kitchen**n**

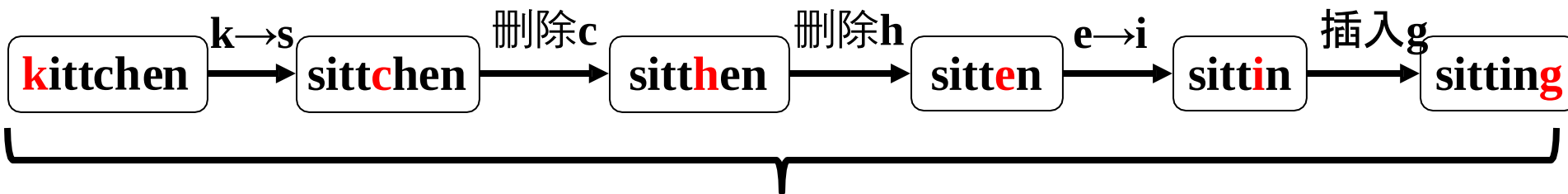
6次

方案1



5次

方案2



问题：如何求出最少的编辑操作数（最小编辑距离）？

- 输入：长度为 n 的字符串 s ，长度为 m 的字符串 t
- 输出： 求出一组编辑操作 $O = \langle e_1, e_2, \dots, e_d \rangle$ ， 令

$$\min |O|$$

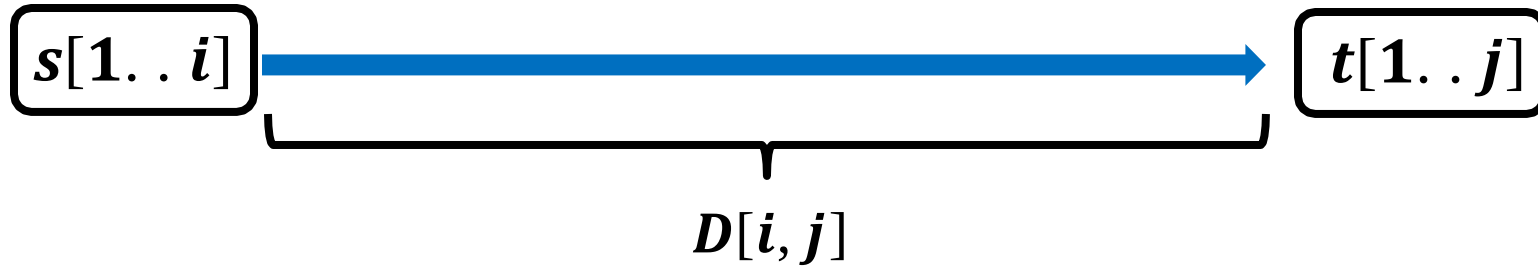
优化目标

$s.t.$ 字符串 s 经过 O 的操作后满足 $s = t$

约束条件

问题结构分析

- 给出问题表示
- $D[i, j]$: 字符串 $s[1..i]$ 变为 $t[1..j]$ 的最小编辑距离



- 明确原始问题
 - $D[n, m]$: 字符串 $s[1..n]$ 变为 $t[1..m]$ 的最小编辑距离

问题结构分析



递推关系建立



自底向上计算



最优方案追踪

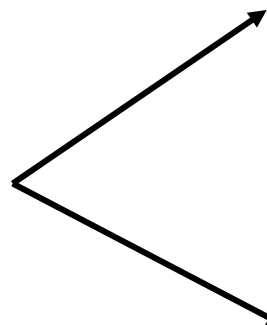
递推关系建立：回顾与启发

- 最长公共子序列

- 如果 $s_i \neq t_j$

<i>s</i>	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

<i>t</i>	B	D	C	A	B	A
----------	---	---	---	---	---	---



<i>s</i>	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

<i>t</i>	B	D	C	A	B	A
----------	---	---	---	---	---	---

<i>s</i>	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

<i>t</i>	B	D	C	A	B	A
----------	---	---	---	---	---	---

- 如果 $s_i = t_j$

<i>s</i>	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

<i>t</i>	B	D	C	A	B
----------	---	---	---	---	---



<i>s</i>	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	---

<i>t</i>	B	D	C	A	B
----------	---	---	---	---	---

考察末尾元素

递推关系建立

- 考察末尾元素

- 删除

<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------

<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------



?

- 插入

<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>	?
----------	----------	----------	----------	----------	----------	----------	----------	---

<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------



?

- 替换

							?
<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>

<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------



?

问题结构分析

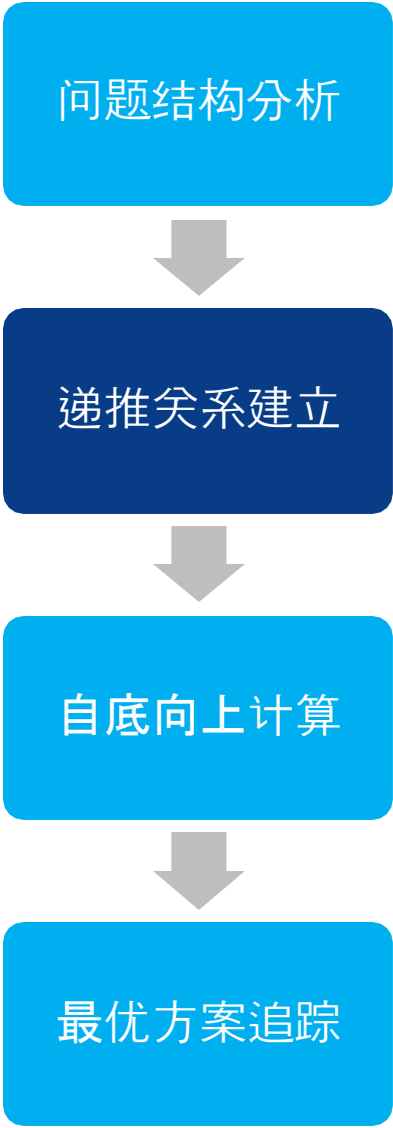
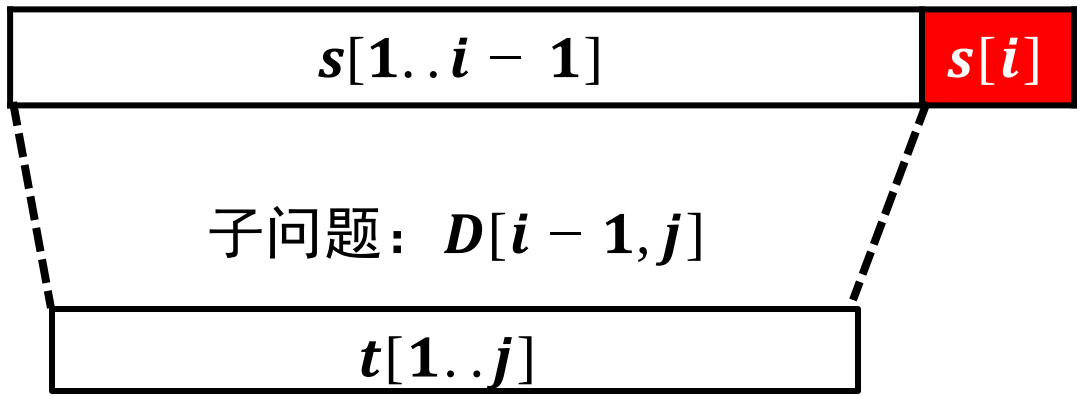
递推关系建立

自底向上计算

最优方案追踪

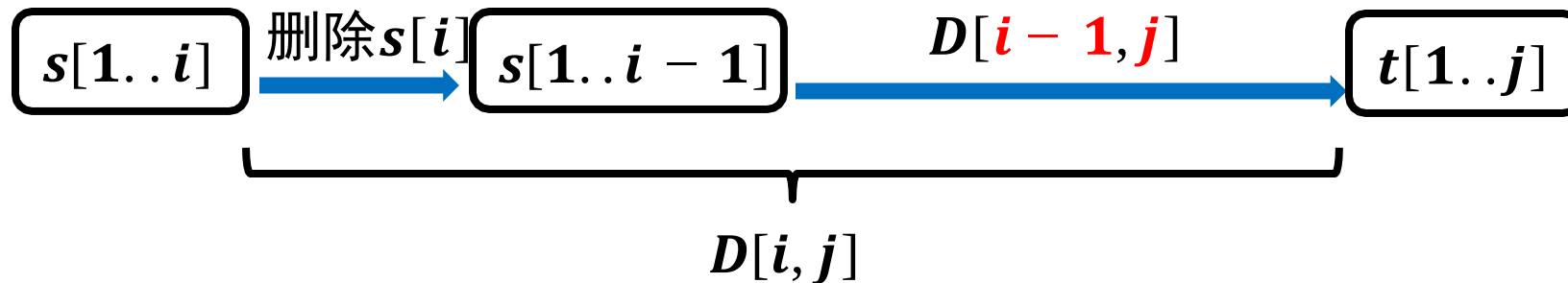
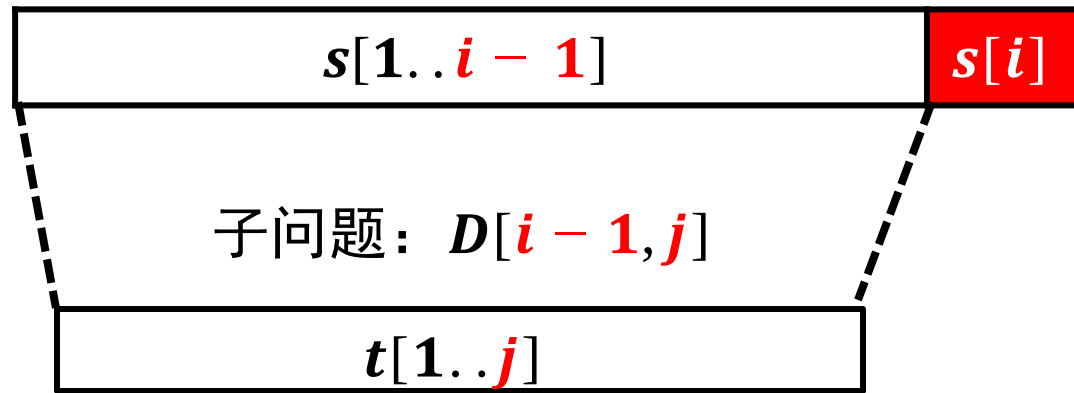
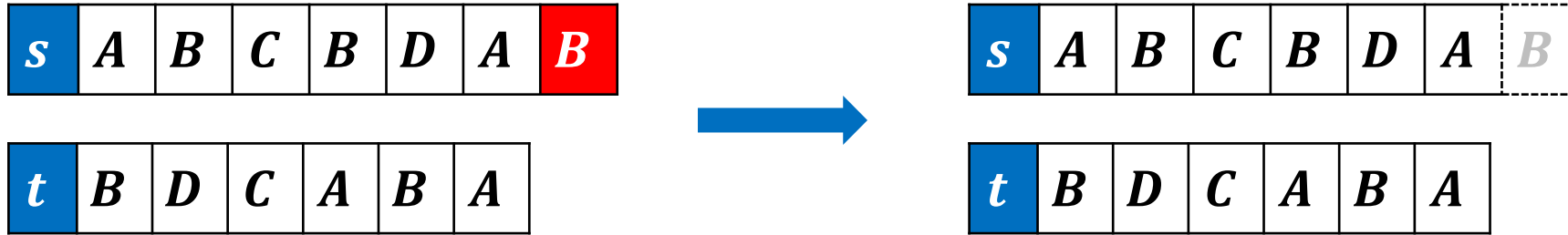
递推关系建立：分析最优（子）结构

- 考察末尾元素：删除



递推关系建立：分析最优（子）结构

- 考察末尾元素：删除



问题结构分析

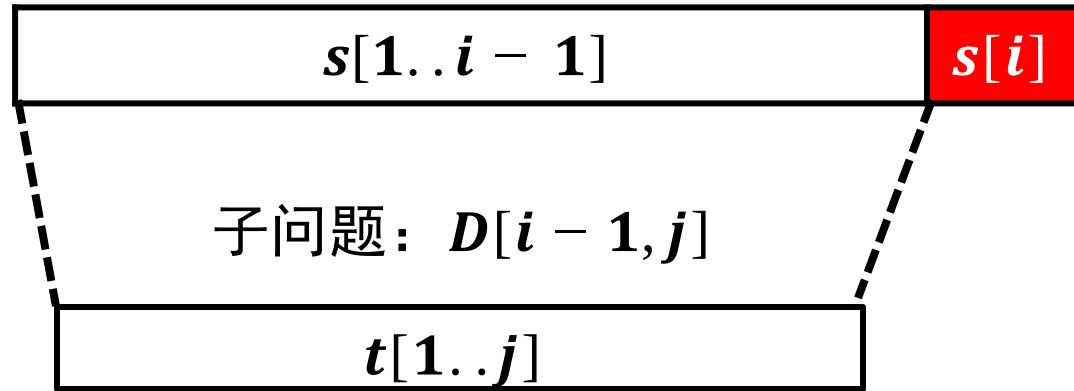
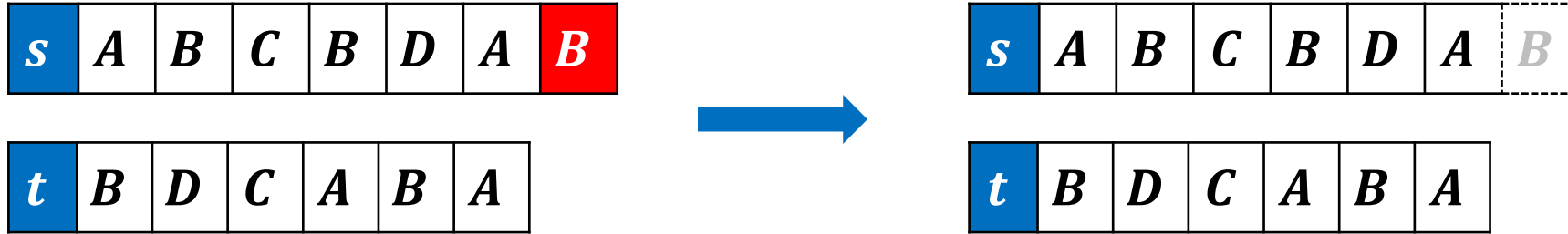
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：删除



$$D[i, j] = D[i-1, j] + 1$$

最优子结构

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：插入

<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>	?
----------	----------	----------	----------	----------	----------	----------	----------	---



<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

问题结构分析



递推关系建立



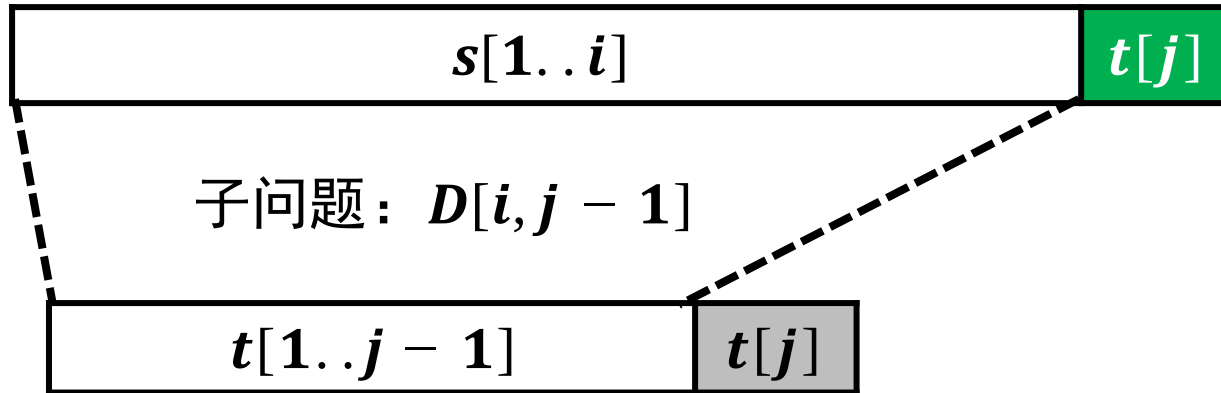
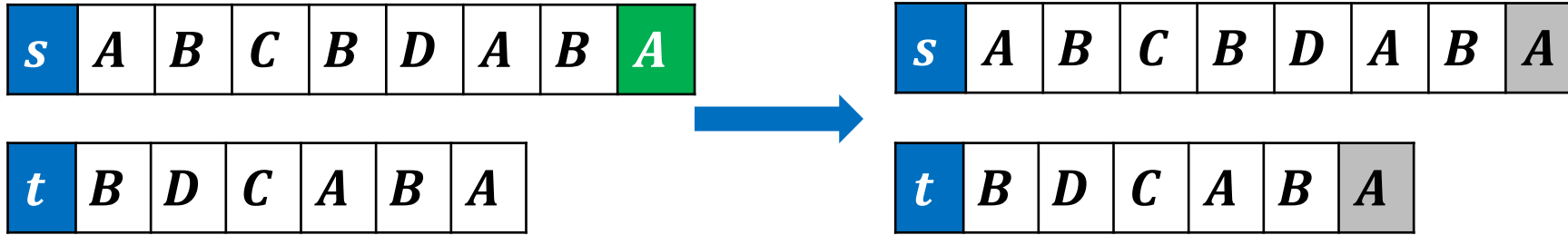
自底向上计算



最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：插入



问题结构分析

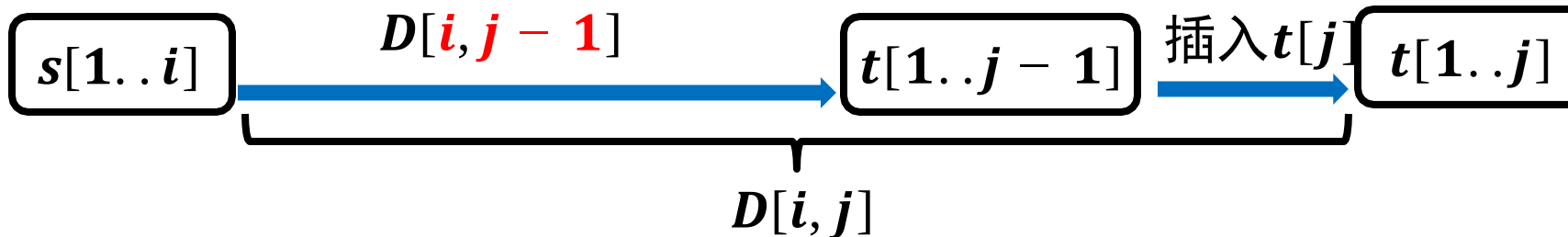
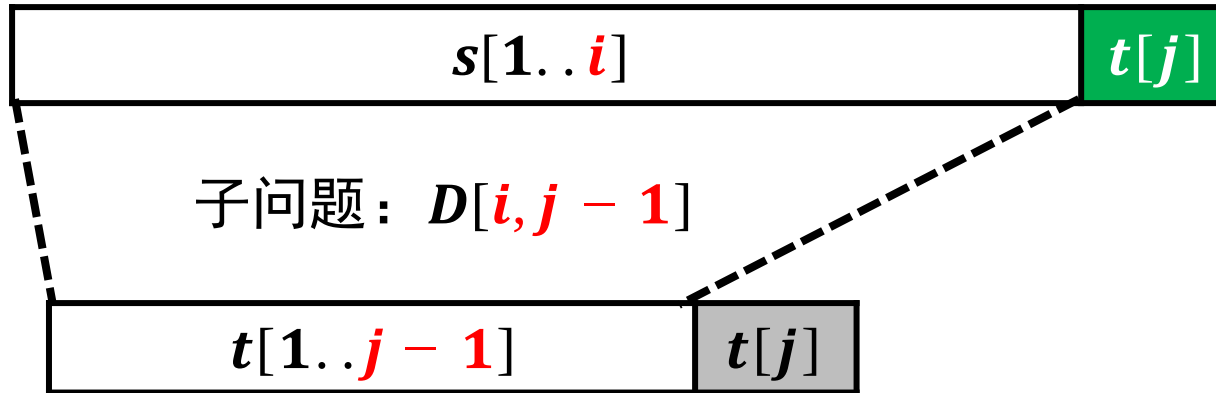
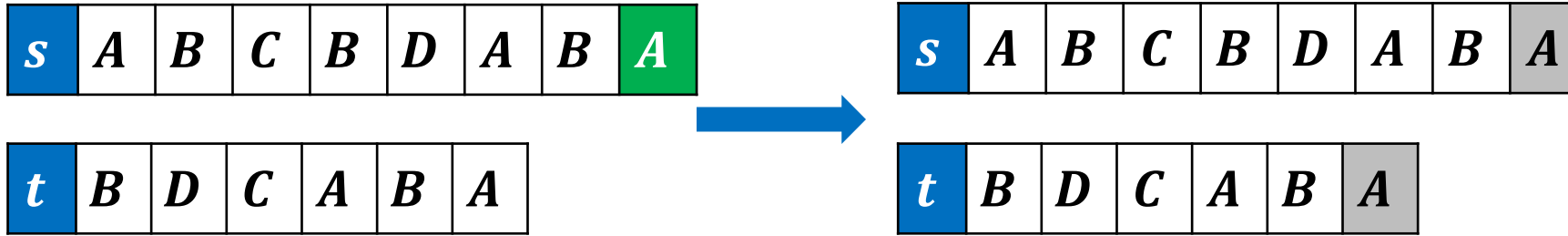
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：插入



问题结构分析

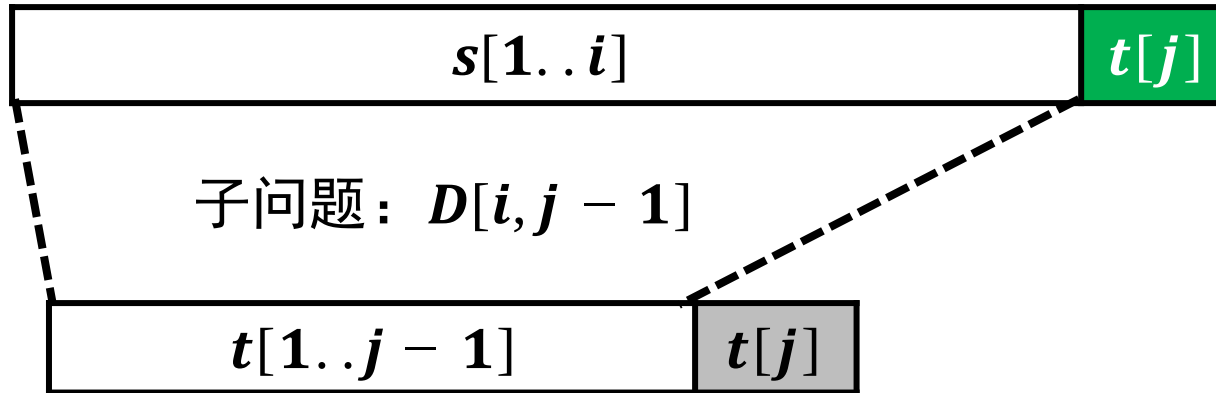
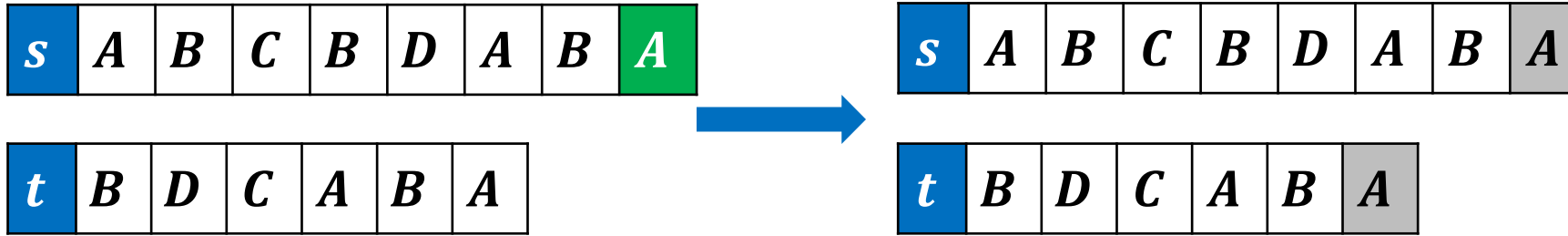
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：插入



$$D[i, j] = D[i, j-1] + 1$$

最优子结构

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：替换

<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	<i>B</i>
----------	----------	----------	----------	----------	----------	----------	----------



<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

问题结构分析



递推关系建立



自底向上计算



最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：替换

<i>s</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>B</i>	<i>D</i>	<i>A</i>	?
----------	----------	----------	----------	----------	----------	----------	---



<i>t</i>	<i>B</i>	<i>D</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>A</i>
----------	----------	----------	----------	----------	----------	----------

问题结构分析



递推关系建立



自底向上计算



最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：替换

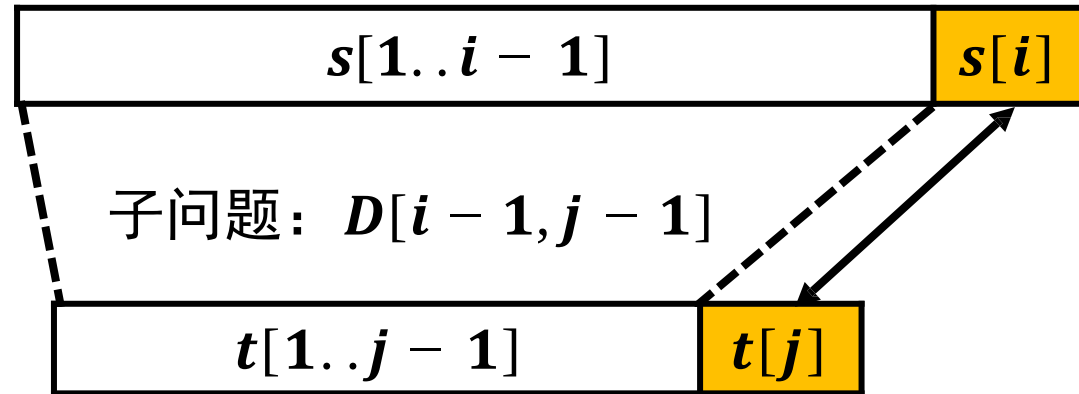
s A B C B D A **A**

t B D C A B **A**



s A B C B D A A

t B D C A B A



问题结构分析



递推关系建立



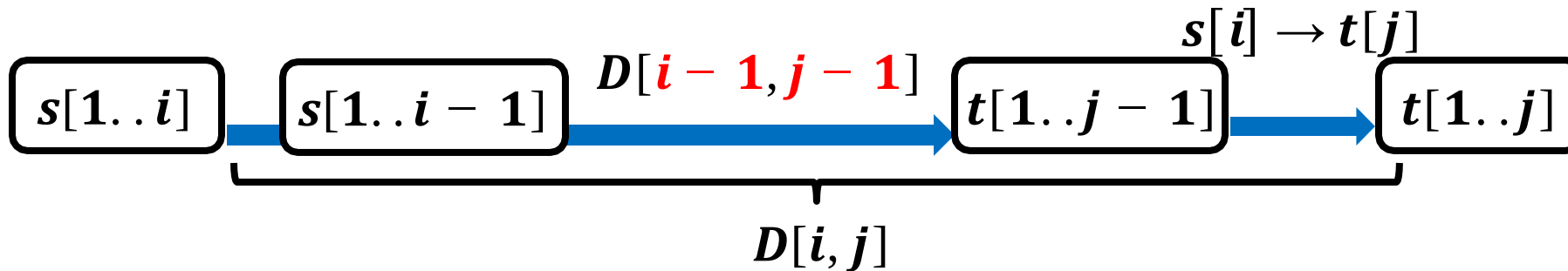
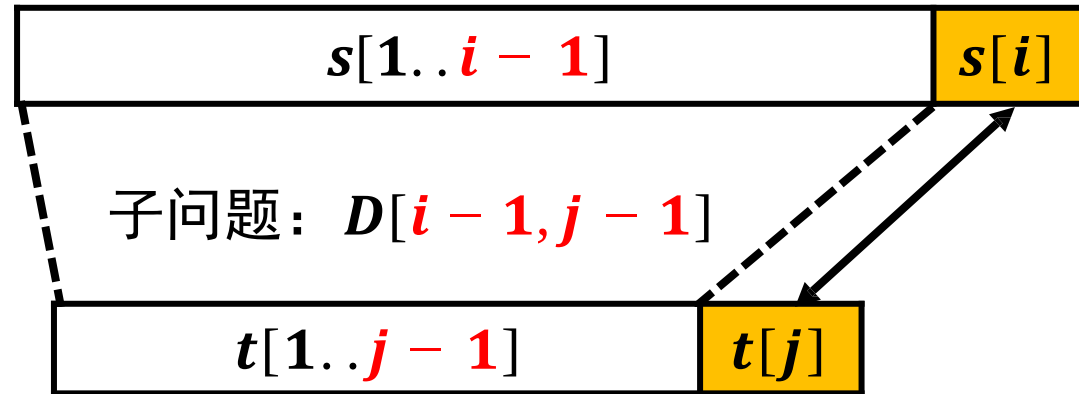
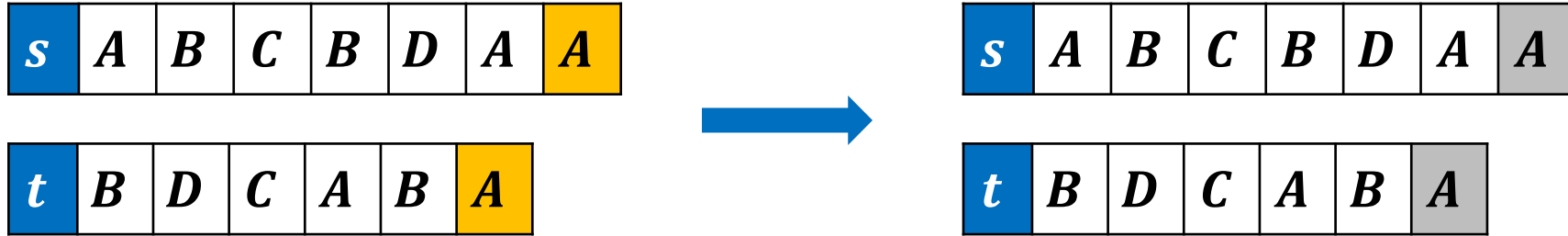
自底向上计算



最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：替换



问题结构分析

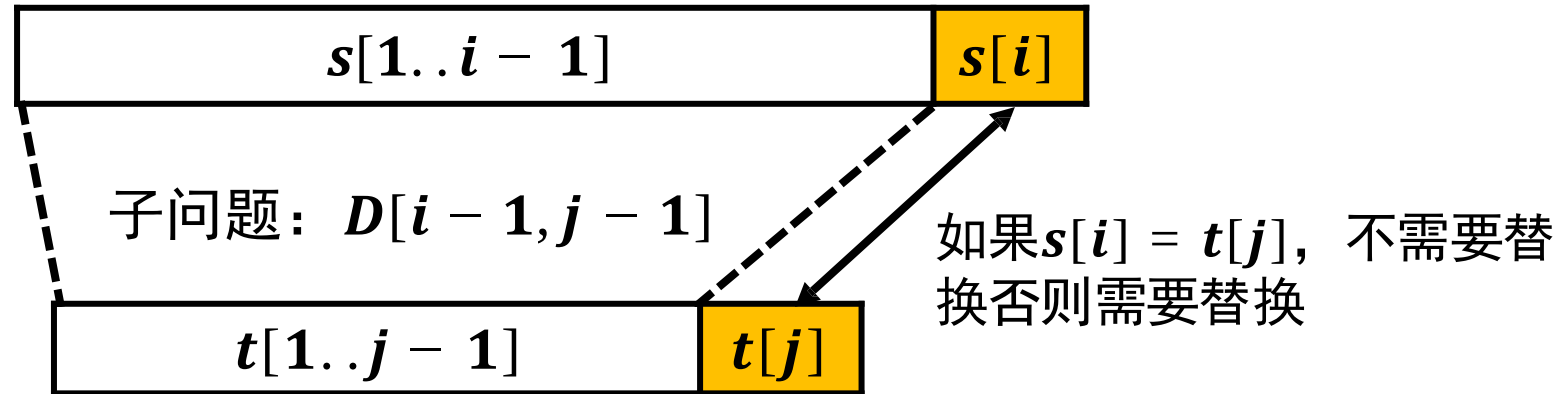
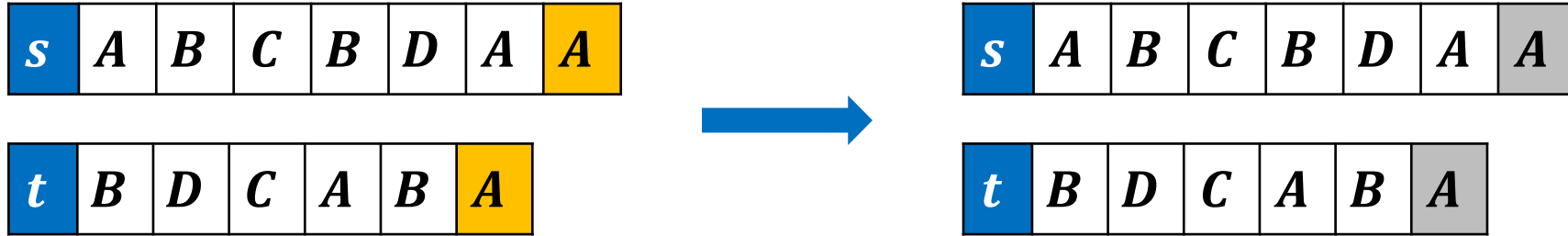
递推关系建立

自底向上计算

最优方案追踪

递推关系建立：分析最优（子）结构

- 考察末尾元素：替换



$$D[i, j] = D[i-1, j-1] + \begin{cases} 0, & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases}$$

最优子结构

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

递推关系建立：构造递推公式

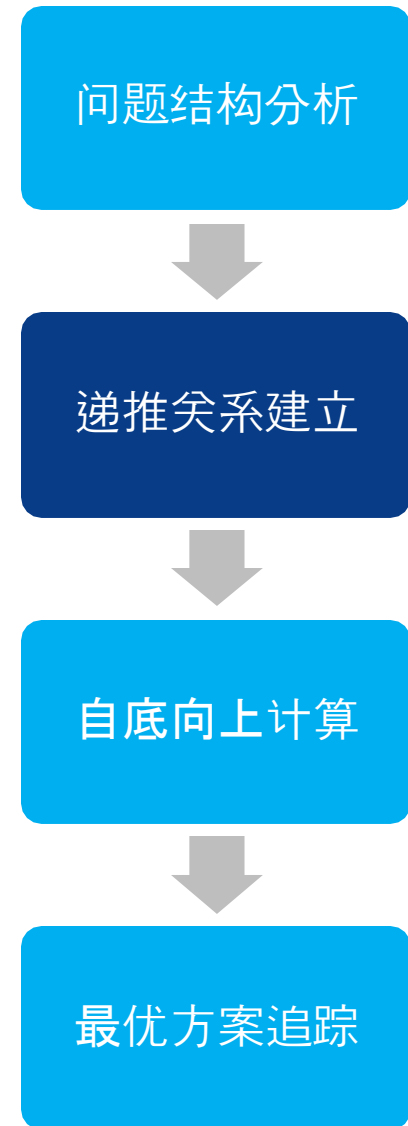
- 综合上面三种方式

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0, & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

删除
插入
替换

- vs. 最长公共子序列

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_i \\ C[i-1, j-1] + 1 & x_i = y_i \end{cases}$$



自底向上计算：确定计算顺序

- 初始化
 - $D[i, 0] = i$
- 把长度为 i 的串变为空串至少需要 i 次操作（删除）
 - $D[0, j] = j$
- 把空串变为长度为 j 的串至少需要 j 次操作（插入）

$i \backslash j$	0	1	2	...	$m - 1$	m
0	0	1	2	...	$m - 1$	m
1	1					
2	2					
...	...					
$n - 1$	$n - 1$					
n	n					

问题结构分析



递推关系建立



自底向上计算



最优方案追踪

自底向上计算：确定计算顺序

• 递推公式

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0, & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

删除
插入
替换

$i \backslash j$	0	1	2	...	$m-1$	m
0	0	1	2	...	$m-1$	m
1	1	$D[i-1, j-1] + 0/1$				
2	2				$D[i-1, j] + 1$	
...	...					
$n-1$	$n-1$					
n	n					

Diagram illustrating the dynamic programming table structure and the calculation of $D[i, j]$ from its neighbors:

- From $D[i-1, j-1]$ (diagonal arrow)
- From $D[i-1, j]$ (vertical arrow)
- From $D[i, j-1]$ (horizontal arrow)

问题结构分析



递推关系建立



自底向上计算



最优方案追踪

自底向上计算：依次计算问题

• 递推公式

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0, & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

删除
插入
替换

$i \backslash j$	0	1	2	...	$m-1$	m
0	0	1	2	...	$m-1$	m
1	1					
2	2					
...	...					
$n-1$	$n-1$					
n	n					

Diagram illustrating the dynamic programming table for sequence alignment. The table shows the distance $D[i, j]$ between the first i characters of sequence s and the first j characters of sequence t . The table is filled with values from 0 to m . Arrows indicate the sequence of calculations: solid arrows point from $(i-1, j)$ to (i, j) (deletion) and from $(i, j-1)$ to (i, j) (insertion). Dashed arrows point from $(i-1, j-1)$ to (i, j) (replacement). A red star marks the final cell (n, m) .

问题结构分析

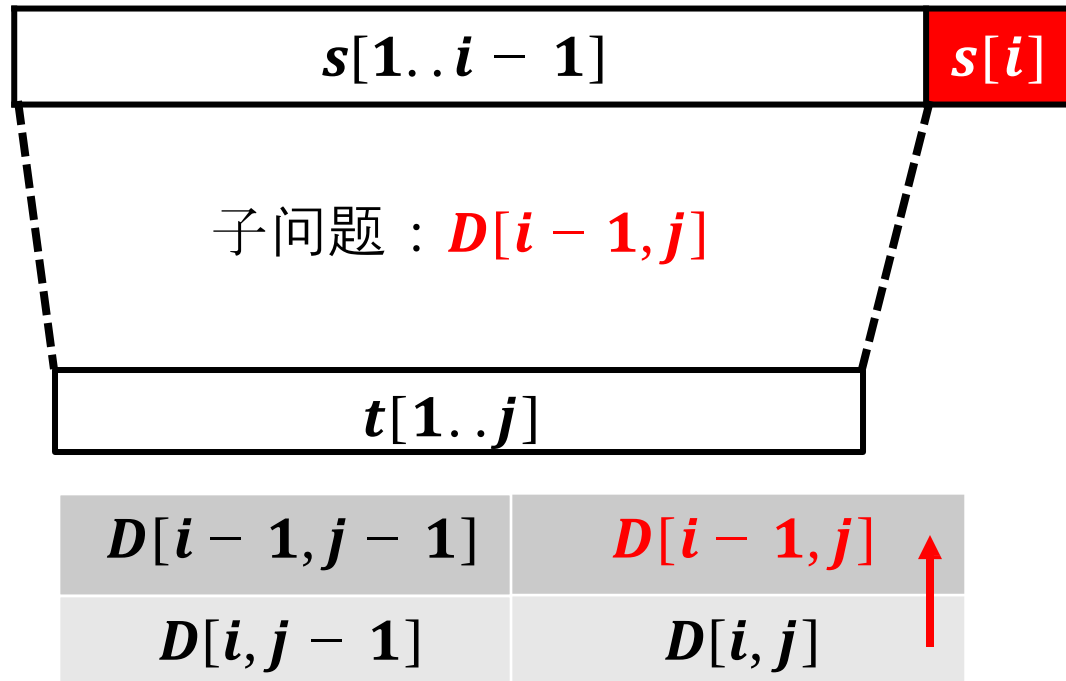
递推关系建立

自底向上计算

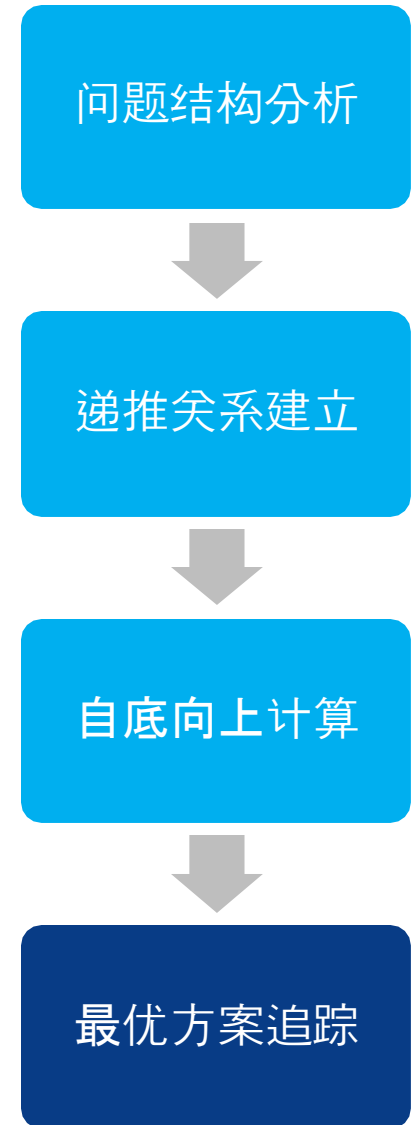
最优方案追踪

最优方案追踪：记录决策过程

- 追踪数组 Rec ，记录子问题来源

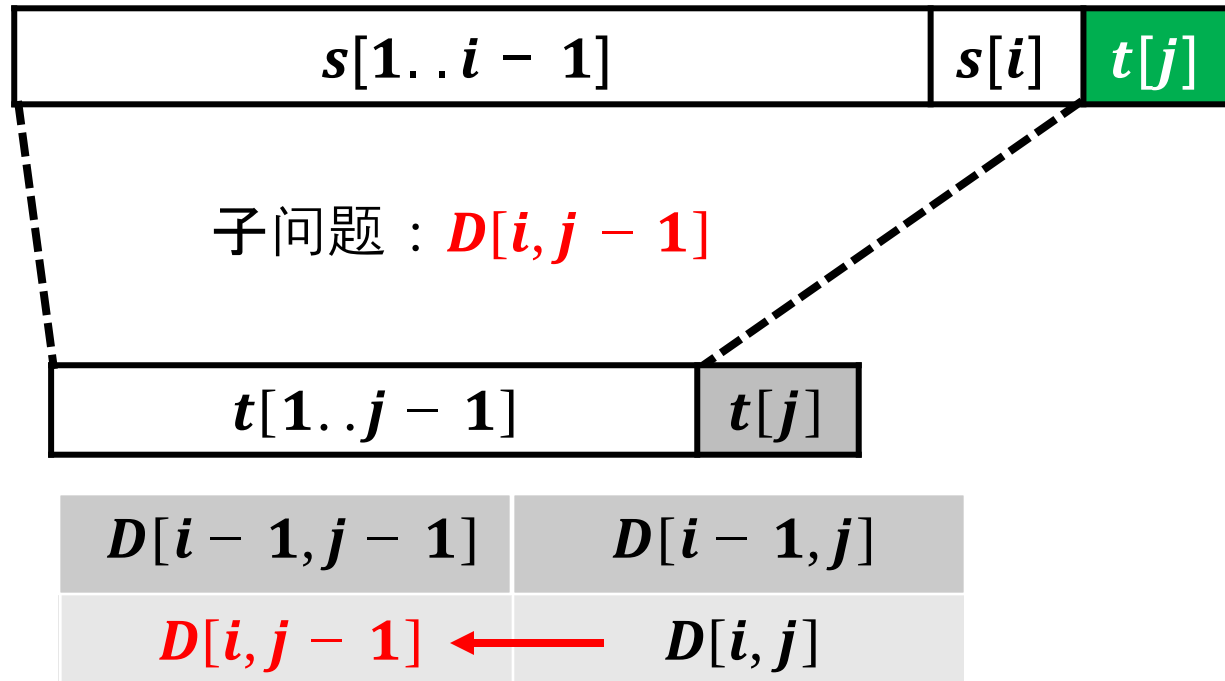


$Rec[i, j]$	子问题来源	操作
U	上侧, 即 $D[i-1, j]$	删除 $s[i]$



最优方案追踪：记录决策过程

- 追踪数组 Rec ，记录子问题来源



$Rec[i, j]$	子问题来源	操作
U	上侧，即 $D[i - 1, j]$	删除 $s[i]$
L	左侧，即 $D[i, j - 1]$	插入 $t[j]$

问题结构分析

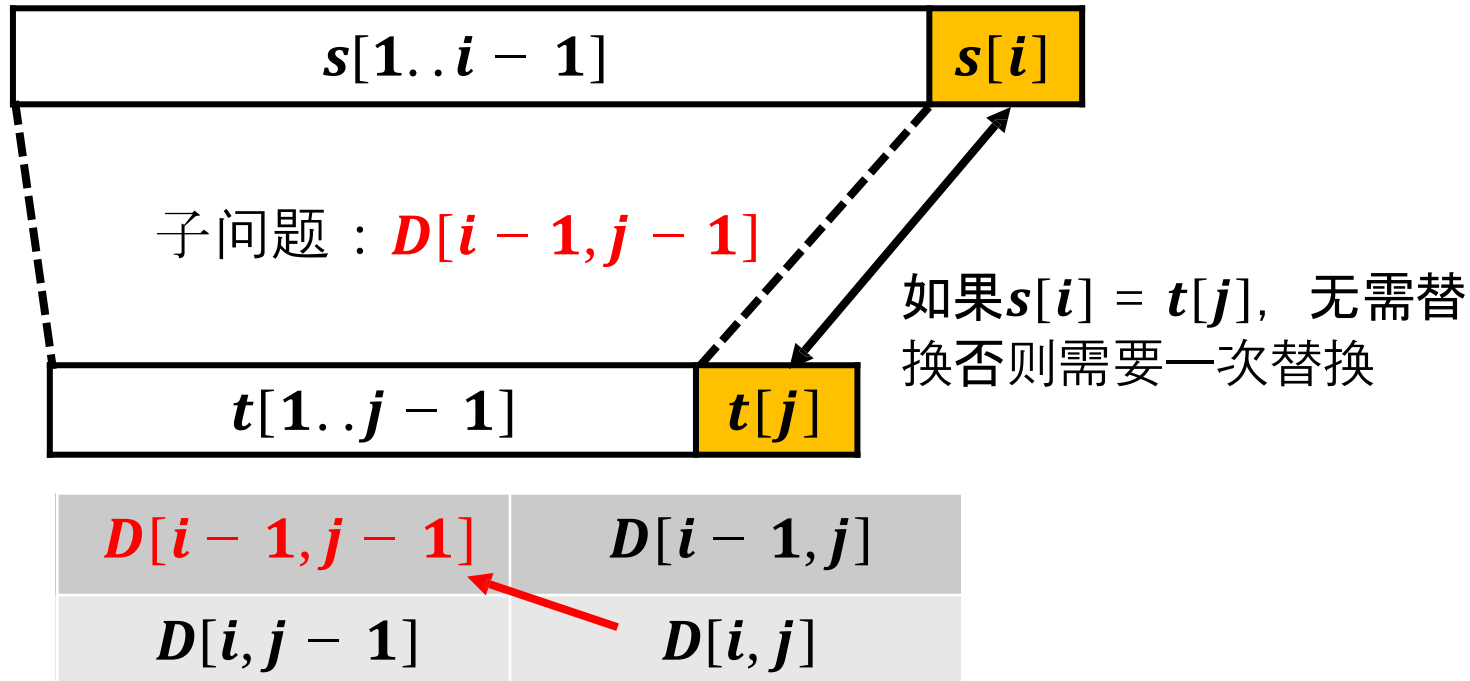
递推关系建立

自底向上计算

最优方案追踪

最优方案追踪：记录决策过程

- 追踪数组 Rec ，记录子问题来源



$Rec[i, j]$	子问题来源	操作
U	上侧, 即 $D[i-1, j]$	删除 $s[i]$
L	左侧, 即 $D[i, j-1]$	插入 $t[j]$
LU	左上, 即 $D[i-1, j-1]$	用 $t[j]$ 替换 $s[i]$ /无操作

问题结构分析

递推关系建立

自底向上计算

最优方案追踪

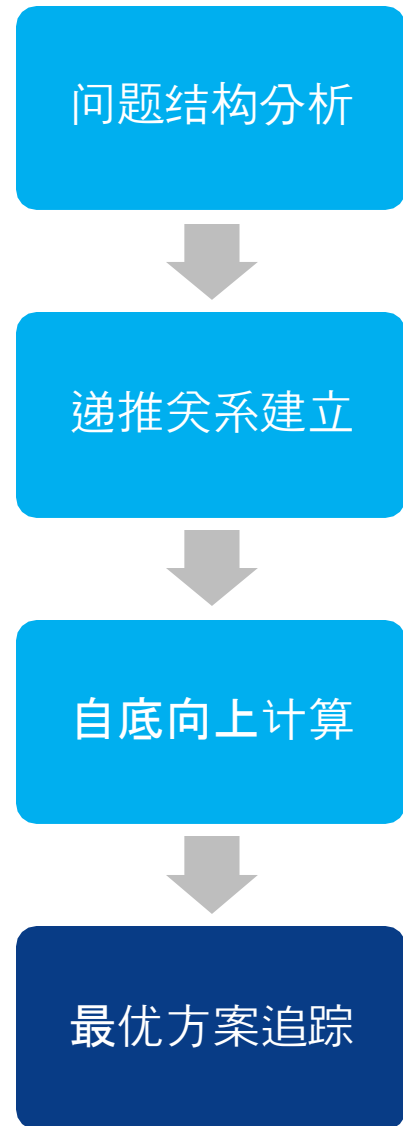
最优方案追踪：输出最优方案

- 根据数组 Rec ，输出最少编辑操作

$i \backslash j$	0	1	2	...	$m-1$	m
0						
1						
2						
...						
$n-1$						
n						

$Rec[] = LU$
 $Rec[] = L$
 $Rec[] = U$

$Rec[i, j]$	子问题来源	操作
U	上侧，即 $D[i-1, j]$	删除 $s[i]$
L	左侧，即 $D[i, j-1]$	插入 $t[j]$
LU	左上，即 $D[i-1, j-1]$	用 $t[j]$ 替换 $s[i]$ /无操作



	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1						
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

初始化

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U						
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1						
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U						
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 & \text{删除A} \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

		$D[i - 1, j]$						
$i \backslash j$	0	1	2	3	4	5	6	
0	0	1	2	3	4	5	6	
1	1							
2	2							
3	3							
4	4							
5	5							
6	6							
7	7							

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U						
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 & \text{删除A} \\ D[i, j-1] + 1 & \text{插入B} \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1						
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

$D[i-1, j]$

$D[i, j-1]$

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U						
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
s	A	B	C	B	D	A	B
t	B	D				A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 & \text{删除A} \\ D[i, j-1] + 1 & \text{插入B} \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} & \text{A替换为B} \end{cases}$$

D

$i \backslash j$	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1						
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

$i \backslash j$	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U						
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D				A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 & \text{删除A} \\ D[i, j-1] + 1 & \text{插入B} \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} & \text{A替换为B} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1					
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU					
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C				

$s[i] \neq t[j]$

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 & \text{删除A} \\ D[i, j-1] + 1 & \text{插入D} \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} & \text{A替换为D} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2				
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU				
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3			
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU			
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	D
<i>t</i>	B	D	C	A	D	A	

$s[i] = t[j]$

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D $D[i-1, j]$ ec

无需替换

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3		
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

$D[i-1, j-1]$

$D[i, j-1]$

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU		
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2						
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U						
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1					
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU					
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2				
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU				
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3			
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU			
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4		
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU		
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3						
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U						
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2					
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U					
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2				
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU				
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2			
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU			
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{cases}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3		
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L		
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	4
4	4						
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U						
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	4
4	4	3					
5	5						
6	6						
7	7						

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU					
5	U						
6	U						
7	U						

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	4
4	4	3	3	3	3	3	4
5	5	4	3	4	4	4	4
6	6	5	4	4	4	5	4
7	7	6	5	5	5	4	

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	4
4	4	3	3	3	3	3	4
5	5	4	3	4	4	4	4
6	6	5	4	4	4	5	4
7	7	6	5	5	5	4	5

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

D

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	1	2	3	3	4	5
2	2	1	2	3	4	3	4
3	3	2	2	2	3	4	4
4	4	3	3	3	3	3	4
5	5	4	3				4
6	6	5	4	4	4	5	4
7	7	6	5	5	5	4	5

最优解

Rec

<i>i</i> \ <i>j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i,j] = \min \left\{ \begin{array}{l} D[i-1,j] + 1 \\ D[i,j-1] + 1 \\ D[i-1,j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i,j] = \min \left\{ \begin{array}{l} D[i-1,j] + 1 \\ D[i,j-1] + 1 \\ D[i-1,j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

无需操作
插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i,j] = \min \left\{ \begin{array}{l} D[i-1,j] + 1 \\ D[i,j-1] + 1 \\ D[i-1,j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

无需操作
无需操作
插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作：

用C替换D
无需操作
无需操作
插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

用D替换B
用C替换D
无需操作
无需操作
插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

删除C

用D替换B

用C替换D

无需操作

无需操作

插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作:

无需操作
删除C
用D替换B
用C替换D
无需操作
无需操作
插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

	1	2	3	4	5	6	7
<i>s</i>	A	B	C	B	D	A	B
<i>t</i>	B	D	C	A	B	A	

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$

操作：
 删除A
 无需操作
 删除C
 用D替换B
 用C替换D
 无需操作
 无需操作
 插入A

Rec

<i>i \ j</i>	0	1	2	3	4	5	6
0		L	L	L	L	L	L
1	U	LU	LU	LU	LU	L	LU
2	U	LU	LU	LU	LU	LU	L
3	U	U	LU	LU	L	L	LU
4	U	LU	LU	LU	LU	LU	L
5	U	U	LU	LU	LU	LU	LU
6	U	U	U	LU	LU	LU	LU
7	U	LU	U	LU	LU	LU	L

- Minimum-Edit-Distance(s, t)

输入: 字符串 s 和 t

输出: s 和 t 的最小编辑距离

$n \leftarrow \text{length}(s)$

$m \leftarrow \text{length}(t)$

新建 $D[0..n, 0..m]$, $Rec[0..n, 0..m]$ 两个二维数组

//初始化

```
for  $i \leftarrow 0$  to  $n$  do
|    $D[i, 0] \leftarrow i$ 
|    $Rec[i, 0] \leftarrow "U"$ 
end
for  $j \leftarrow 0$  to  $m$  do
|    $D[0, j] \leftarrow j$ 
|    $Rec[0, j] \leftarrow "L"$ 
end
```

初始化

- Minimum-Edit-Distance(s, t)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
       $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i-1, j-1] + c$ 
     $delete \leftarrow D[i-1, j] + 1$ 
     $insert \leftarrow D[i, j-1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i-1, j-1] + c$ 
       $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i, j-1] + 1$ 
       $Rec[i, j] \leftarrow "L"$ 
    end
    else
       $D[i, j] \leftarrow D[i-1, j] + 1$ 
       $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

- Minimum-Edit-Distance(s, t)

//动态规划

for $i \leftarrow 1$ to n do

 for $j \leftarrow 1$ to m do

$c \leftarrow 0$

 if $s_i \neq t_j$ then

$c \leftarrow 1$

 end

$replace \leftarrow D[i-1, j-1] + c$

$delete \leftarrow D[i-1, j] + 1$

$insert \leftarrow D[i, j-1] + 1$

 if $replace = \min\{delete, insert, replace\}$ then

$D[i, j] \leftarrow D[i-1, j-1] + c$

$Rec[i, j] \leftarrow "LU"$

 end

 else if $insert = \min\{delete, insert, replace\}$ then

$D[i, j] \leftarrow D[i, j-1] + 1$

$Rec[i, j] \leftarrow "L"$

 end

 else

$D[i, j] \leftarrow D[i-1, j] + 1$

$Rec[i, j] \leftarrow "U"$

 end

 end

end

替换/无需操作

- Minimum-Edit-Distance(s, t)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
       $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i-1, j-1] + c$ 
     $delete \leftarrow D[i-1, j] + 1$ 
     $insert \leftarrow D[i, j-1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i-1, j-1] + c$ 
       $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i, j-1] + 1$ 
       $Rec[i, j] \leftarrow "L"$ 
    end
    else
       $D[i, j] \leftarrow D[i-1, j] + 1$ 
       $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

采用替换操作

- Minimum-Edit-Distance(s, t)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
      |  $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i - 1, j - 1] + c$ 
     $delete \leftarrow D[i - 1, j] + 1$ 
     $insert \leftarrow D[i, j - 1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
      |  $D[i, j] \leftarrow D[i - 1, j - 1] + c$ 
      |  $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
      |  $D[i, j] \leftarrow D[i, j - 1] + 1$ 
      |  $Rec[i, j] \leftarrow "L"$ 
    end
    else
      |  $D[i, j] \leftarrow D[i - 1, j] + 1$ 
      |  $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

记录编辑距离和操作

- Minimum-Edit-Distance(s, t)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
       $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i-1, j-1] + c$ 
     $delete \leftarrow D[i-1, j] + 1$ 
     $insert \leftarrow D[i, j-1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i-1, j-1] + c$ 
       $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i, j-1] + 1$ 
       $Rec[i, j] \leftarrow "L"$ 
    end
    else
       $D[i, j] \leftarrow D[i-1, j] + 1$ 
       $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

采取插入操作

- Minimum-Edit-Distance(s, t)

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
      |  $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i-1, j-1] + c$ 
     $delete \leftarrow D[i-1, j] + 1$ 
     $insert \leftarrow D[i, j-1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
      |  $D[i, j] \leftarrow D[i-1, j-1] + c$ 
      |  $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
      |  $D[i, j] \leftarrow D[i, j-1] + 1$ 
      |  $Rec[i, j] \leftarrow "L"$ 
    end
    else
      |  $D[i, j] \leftarrow D[i-1, j] + 1$ 
      |  $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

采取删除操作

- **Print-MED** (Rec, s, t, i, j)

输入: 矩阵 Rec , 字符串 s, t , 位置索引 i 和 j

输出: 操作序列

if $i = 0$ and $j = 0$ then

| return $NULL$

end

if $Rec[i, j] = "LU"$ then

| Print-MED($Rec, s, t, i - 1, j - 1$)

| if $s_i = t_j$ then

| | print “无需操作”

| end

| else

| | print “用 t_j 替换 s_i ”

| end

end

else if $Rec[i, j] = "U"$ then

| Print-MED($Rec, s, t, i - 1, j$)

| print “删除 s_i ”

end

else

| Print-MED($Rec, s, t, i, j - 1$)

| print “插入 t_j ”

end

采取替换操作

- **Print-MED**(Rec, s, t, i, j)

输入: 矩阵 Rec , 字符串 s, t , 位置索引 i 和 j

输出: 操作序列

if $i = 0$ and $j = 0$ then

 | return $NULL$

end

if $Rec[i, j] = "LU"$ then

 | Print-MED($Rec, s, t, i - 1, j - 1$)

 | if $s_i = t_j$ then

 | print “无需操作”

 end

 else

 | print “用 t_j 替换 s_i ”

 end

end

else if $Rec[i, j] = "U"$ then

 | Print-MED($Rec, s, t, i - 1, j$)

 | print “删除 s_i ”

end

else

 | Print-MED($Rec, s, t, i, j - 1$)

 | print “插入 t_j ”

end

递归输出子问题方案

- **Print-MED**(Rec, s, t, i, j)

输入: 矩阵 Rec , 字符串 s, t , 位置索引 i 和 j

输出: 操作序列

if $i = 0$ and $j = 0$ then

 | return $NULL$

end

if $Rec[i, j] = "LU"$ then

 | Print-MED($Rec, s, t, i - 1, j - 1$)

 if $s_i = t_j$ then

 | print “无需操作”

 end

 else

 | print “用 t_j 替换 s_i ”

 end

end

else if $Rec[i, j] = "U"$ then

 | Print-MED($Rec, s, t, i - 1, j$)

 | print “删除 s_i ”

end

else

 | Print-MED($Rec, s, t, i, j - 1$)

 | print “插入 t_j ”

end

替换/无操作

- **Print-MED** (Rec, s, t, i, j)

输入: 矩阵 Rec , 字符串 s, t , 位置索引 i 和 j

输出: 操作序列

if $i = 0$ and $j = 0$ then

 | return $NULL$

end

if $Rec[i, j] = "LU"$ then

 | Print-MED($Rec, s, t, i - 1, j - 1$)

 | if $s_i = t_j$ then

 | print “无需操作”

 end

 else

 | print “用 t_j 替换 s_i ”

 end

end

else if $Rec[i, j] = "U"$ then

 | Print-MED($Rec, s, t, i - 1, j$)

 | print “删除 s_i ”

end

else

 | Print-MED($Rec, s, t, i, j - 1$)

 | print “插入 t_j ”

end

采取删除操作

- **Print-MED**(Rec, s, t, i, j)

输入: 矩阵 Rec , 字符串 s, t , 位置索引 i 和 j

输出: 操作序列

if $i = 0$ and $j = 0$ then

 | return $NULL$

end

if $Rec[i, j] = "LU"$ then

 | Print-MED($Rec, s, t, i - 1, j - 1$)

 if $s_i = t_j$ then

 | print “无需操作”

 end

 else

 | print “用 t_j 替换 s_i ”

 end

end

else if $Rec[i, j] = "U"$ then

 | Print-MED($Rec, s, t, i - 1, j$)

 | print “删除 s_i ”

end

else

 | Print-MED($Rec, s, t, i, j - 1$)

 | print “插入 t_j ”

end

采取插入操作

//动态规划

```
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $m$  do
     $c \leftarrow 0$ 
    if  $s_i \neq t_j$  then
       $c \leftarrow 1$ 
    end
     $replace \leftarrow D[i-1, j-1] + c$ 
     $delete \leftarrow D[i-1, j] + 1$ 
     $insert \leftarrow D[i, j-1] + 1$ 
    if  $replace = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i-1, j-1] + c$ 
       $Rec[i, j] \leftarrow "LU"$ 
    end
    else if  $insert = \min\{delete, insert, replace\}$  then
       $D[i, j] \leftarrow D[i, j-1] + 1$ 
       $Rec[i, j] \leftarrow "L"$ 
    end
    else
       $D[i, j] \leftarrow D[i-1, j] + 1$ 
       $Rec[i, j] \leftarrow "U"$ 
    end
  end
end
end
```

时间复杂度 : $O(mn)$

$O(m)$

$O(mn)$

最长公共子序列

如果 $s_i \neq t_j$

s	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

t	B	D	C	A	B	A
----------	---	---	---	---	---	----------

如果 $s_i = t_j$

s	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

t	B	D	C	A	B
----------	---	---	---	---	----------

$$C[i, j] = \begin{cases} \max\{C[i-1, j], C[i, j-1]\} & x_i \neq y_j \\ C[i-1, j-1] + 1 & x_i = y_j \end{cases}$$

最小编辑距离

s	A	B	C	B	D	A	B
----------	---	---	---	---	---	---	----------

删除

t	B	D	C	A	B	A
----------	---	---	---	---	---	---

s	A	B	C	B	D	A	B	?
----------	---	---	---	---	---	---	---	----------

插入

t	B	D	C	A	B	A
----------	---	---	---	---	---	---

s	A	B	C	B	D	A	B	?
----------	---	---	---	---	---	---	----------	----------

替换

t	B	D	C	A	B	A
----------	---	---	---	---	---	---

$$D[i, j] = \min \left\{ \begin{array}{l} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \begin{cases} 0 & \text{if } s[i] = t[j] \\ 1 & \text{if } s[i] \neq t[j] \end{cases} \end{array} \right\}$$